

NETWORKWALKS ACADEMY

WEEK 2 — FINAL CYBERSECURITY ASSESSMENT REPORT

Prepared By: Chandrashekar Bala**Program:** Cybersecurity Internship**Organization:** Networkwalks**Assessment:** Week 2 – Cybersecurity Reconnaissance & Network Scanning**Assessment Period:** September 2026

Footprinting • Passive Reconnaissance • GHDB • Maltego Analysis • Nmap • Zenmap

Document Field	Value
Target	networkwalks.com
Resolved IPv4	192.232.216.135
Assessment Window	September 2026
Assessment Type	Authorized external cybersecurity assessment / internship project
Modules Covered	W2-PM1, W2-PM2, W2-PM3, W2-PM4, W2-PM5 / PM-B
Primary Platforms	Kali Linux; Zenmap / Nmap workflow
Evidence Package	52 unique evidence captures plus preserved raw outputs/source material
Reporting Approach	Evidence-first; observation, validation, attribution and impact kept distinct

Submission-ready reporting standard

This document is written as a consolidated technical report rather than a command transcript. Each major activity explains what the tool does, why it was selected, what the captured evidence demonstrates, how an analyst would interpret it, and what should or should not be inferred from the result.

1. Executive Summary

Week 2 established an external attack-surface picture of networkwalks.com by combining administrative intelligence, DNS analysis, web fingerprinting, search-engine reconnaissance methodology, relationship-oriented analysis, Certificate Transparency intelligence, passive subdomain discovery and network-service enumeration.

The assessment resolved networkwalks.com to 192.232.216.135 and identified HostGator-associated authoritative DNS infrastructure. The web layer exposed Apache and WordPress-related technology signals, including WordPress Download Manager behavior and a WordPress REST API reference in the HTTP Link header. WAFW00F identified ModSecurity (SpiderLabs).

Passive certificate analysis produced ten normalized certificate-derived names, while Subfinder independently returned eight of those names. This overlap provides a useful cross-source confidence signal, while historical certificate records were kept separate from claims about current exposure.

Nmap 7.99 provided the network-enumeration layer. A broad top-10,000 TCP scan produced 3,410 reported open TCP states and 4,977 filtered states, but the scan also showed retransmission and timing variability. Targeted service validation was therefore treated as the stronger service-attribution layer. Positively identified services included Pure-FTPd, OpenSSH 9.9, ISC BIND 9.16.23, Apache, Dovecot and Exim 4.100.

The assessment did not establish a CVE, exploitability, authentication bypass, credential compromise, anonymous FTP access, open SMTP relay, DNS zone-transfer weakness, SSH authentication weakness, cPanel/WHM authorization flaw or unauthorized access. The value of the work is the evidence-backed attack-surface map and the disciplined distinction between observations and confirmed security findings.

1.1 Assessment at a Glance

Area	Key result	Security meaning
Domain / DNS	192.232.216.135; HostGator NS; mail host; TXT/SPF data	Defines public infrastructure and trust relationships
Web	Apache + WordPress-related fingerprints	Identifies technologies for later authorized review
WAF	ModSecurity (SpiderLabs)	Confirms a defensive layer; does not prove coverage or bypass resistance
Passive CT	10 normalized names	Provides certificate-derived namespace and historical clues
Subfinder	8 names overlapping CT	Independent corroboration of passive hostname intelligence
Nmap	Large TCP response surface; targeted validation completed	Maps externally reachable service surfaces
Zenmap	GUI-based Nmap configuration/evidence captured	Provides a graphical view of the Nmap workflow
Confirmed vulnerability	None established from these phases	Further authorized validation would be required for vulnerability claims

ASSESSMENT DISCLAIMER

EDUCATIONAL PURPOSES ONLY — This report is prepared solely for authorized cybersecurity education, internship assessment, training, documentation and defensive learning. Any techniques or observations described here must only be applied to systems for which explicit authorization has been provided. The report is not intended to facilitate unauthorized access, exploitation, credential theft, persistence, disruption, or destructive activity.

This report documents security assessment activity performed against Networkwalks / networkwalks.com within the authorized internship assessment scope. Testing was limited to reconnaissance, footprinting, network/service enumeration, web metadata inspection, certificate intelligence, GHDB methodology, Maltego relationship analysis, and Zenmap/Nmap workflows described in this report. Evidence is presented to distinguish directly observed results from analytical interpretation. A visible service, technology fingerprint, certificate name, indexed result, or defensive control identification is not by itself proof of a security vulnerability. No credential theft, unauthorized access, exploitation, persistence, denial of service, or destructive activity is claimed unless explicitly supported by evidence in this report.

2. Scope, Authorization and Evidence Discipline

The work was performed within the authorized Networkwalks internship assessment context. The report covers reconnaissance and enumeration activities and does not present credential attacks, brute force, persistence, destructive testing or denial-of-service activity as part of the evidence set.

A central reporting rule is that a scanner label is not automatically a vulnerability. A port marked open is a network state; a version string is a fingerprint; a certificate name is historical intelligence unless current addressability is independently established; and a WAF fingerprint describes defensive architecture rather than security effectiveness.

Evidence rule

Every conclusion in this report is tied to captured evidence, preserved raw output or supplied project material. Where a result is a candidate rather than a validated service or vulnerability, that distinction is stated explicitly.

3. Week 2 Assessment Methodology

The modules form a practical progression from low-interaction information gathering to controlled network enumeration:



The workflow deliberately uses multiple independent sources. For example, direct authoritative DNS queries were used to validate DNS behavior after DNSRecon timed out, and Subfinder results were compared with certificate-derived names rather than treated as a standalone inventory.

4. Tools Used — Why Each Tool Was Used and What It Produced

Tool	Why it was used	Outcome / evidence value
WHOIS	Registration and delegation context	GoDaddy registrar context, HostGator nameservers, domain status and DNSSEC state
nslookup	Resolver-path and record discovery	A, NS, MX, TXT and SOA observations; public resolution to 192.232.216.135
dig	Independent authoritative DNS validation	Both HostGator nameservers answered for the target; validated DNS observations
DNSRecon	Automated DNS enumeration	Repeated timeout behavior captured; later compared against direct DNS responses
WhatWeb	External web technology fingerprinting	Apache / WordPress / plugin-related fingerprints
cURL	HTTP response and header inspection	HTTP/2 200, REST API Link header, cookie and server metadata
WAFW00F	WAF fingerprinting	ModSecurity (SpiderLabs) identified

SpiderFoot	OSINT workflow and reconnaissance capability	Version 4.0.0, executable/source location and available module/event context
GHDB / search-engine methodology	Search-indexed exposure reconnaissance	Training exercise covering indexed exposed interfaces and PDF listings
Maltego	Relationship-oriented reconnaissance model	Entity/relationship analysis framework used to organize correlation logic
theHarvester 4.10.1	Passive hostname and certificate-source collection	CertSpotter and crt.sh source results
crt.sh	Certificate Transparency collection	Normalized certificate-derived hostname dataset and historical timeline
CertSpotter	Independent certificate-source perspective	Wildcard namespace observation
Subfinder 2.16.0	Passive subdomain discovery	Eight discovered names, overlapping the CT-derived namespace
Nmap 7.99	Network discovery and service validation	Host discovery, TCP enumeration, version validation, NSE, HTTP/TLS and OS evidence
Zenmap 7.99	Graphical Nmap workflow	GUI target/profile configuration and graphical review of the Nmap workflow
Kali Linux	Assessment operating environment	Command-line execution and evidence preservation

5. W2-PM1 — Footprinting & Reconnaissance with Multiple Kali Tools

PM1 established the baseline external footprint. Each tool answered a different question, and the outputs were correlated rather than interpreted independently.

5.1 PM1 Workflow

```

networkwalks.com
├── WHOIS → registration / delegation context
├── nslookup → public DNS records
├── dig → authoritative validation
├── DNSRecon → automated enumeration behavior
├── WhatWeb → web technology fingerprint
├── cURL → HTTP headers / cookies / API references
├── WAFW00F → defensive-layer fingerprint
└── SpiderFoot → OSINT tooling and passive-recon context

```

5.2 Domain Resolution and Registration Context

WHOIS identified GoDaddy.com, LLC as the registrar and delegated DNS to ns6135.hostgator.com and ns6136.hostgator.com. The record also showed client-side domain locks and an unsigned DNSSEC state. These are administrative and configuration observations, not standalone vulnerabilities.

Evidence 01 — DNS resolution evidence

WHY WE PERFORMED THIS: The capture provides the initial target-resolution context used to anchor the subsequent DNS and web checks.

WHAT THIS EVIDENCE SHOWS: Confirms the target hostname-to-IP relationship and provides the network address used for subsequent authorized enumeration.

```
(root@kali)-[~]
# nslookup networkwalks.com
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
```

Evidence 09 — DNS resolution validation

WHY WE PERFORMED THIS: Visual proof retained as part of the Week 2 evidence chain.

WHAT THIS EVIDENCE SHOWS: Confirms the target hostname-to-IP relationship and provides the network address used for subsequent authorized enumeration.

```
(root@kali)-[~]
# nslookup networkwalks.com
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out

# nslookup -type=NS networkwalks.com
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
Server:      8.8.8.8
Address:     8.8.8.8#53

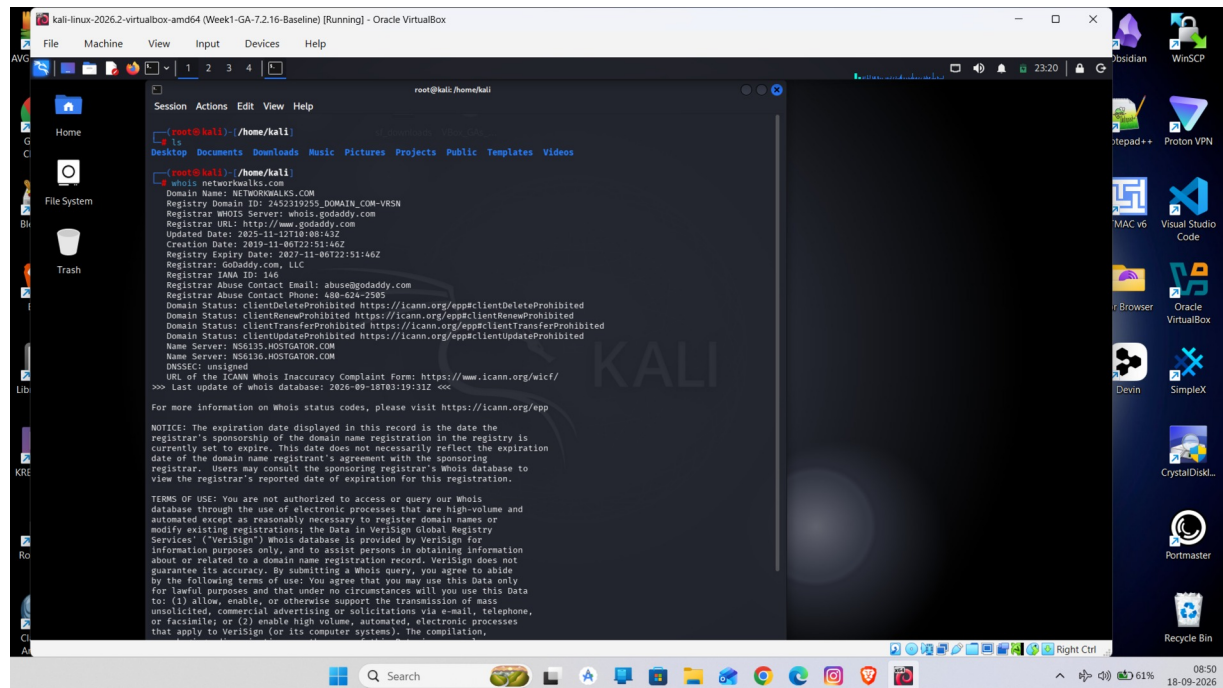
Non-authoritative answer:
networkwalks.com      nameserver = ns6136.hostgator.com.
networkwalks.com      nameserver = ns6135.hostgator.com.

Authoritative answers can be found from:
```

Evidence 47 — WHOIS registration and delegation evidence

WHY WE PERFORMED THIS: The capture records registrar, nameserver and domain-status information used to establish the public administrative/infrastructure context.

WHAT THIS EVIDENCE SHOWS: Shows public registration, delegation and domain-status information used to establish administrative and infrastructure context.



The screenshot shows a Kali Linux desktop environment with a terminal window open. The terminal displays the output of a WHOIS query for the domain networkwalks.com. The output includes domain name, registry ID, registrar information, creation and expiration dates, and various status codes. The terminal window is titled 'kali-linux-2026.2-virtualbox-amd64 (Week1-GA-7.2.16-Baseline) [Running] - Oracle VirtualBox'. The desktop background is the Kali Linux logo, and the taskbar at the bottom shows various application icons.

```
root@kali:~/home/kali# whois networkwalks.com
Domain Name: NETWORKWALKS.COM
Registry Domain ID: 2452319255_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.godaddy.com
Registrar URL: http://www.godaddy.com
Updated Date: 2025-11-12T18:08:43Z
Creation Date: 2019-11-06T22:51:46Z
Registry Expiry Date: 2027-11-06T22:51:46Z
Registrar: GoDaddy.com, LLC
Registrar IANA ID: 146
Registrar Abuse Contact Email: abuse@godaddy.com
Registrar Abuse Contact Phone: 480-624-2505
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientRenewProhibited https://icann.org/epp#clientRenewProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Name Server: NS6135.HOSTGATOR.COM
Name Server: NS6136.HOSTGATOR.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-09-18T03:19:31Z<<<

For more information on WHOIS status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's WHOIS database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our WHOIS
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the data in VeriSign Global Registry
Services' ("VeriSign") WHOIS database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a WHOIS query, you agree to abide
by the following terms of use: You agree that you may use this data only
for lawful purposes and that under no circumstances will you use this data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
```

Evidence 04 — WHOIS / PM1 preservation evidence

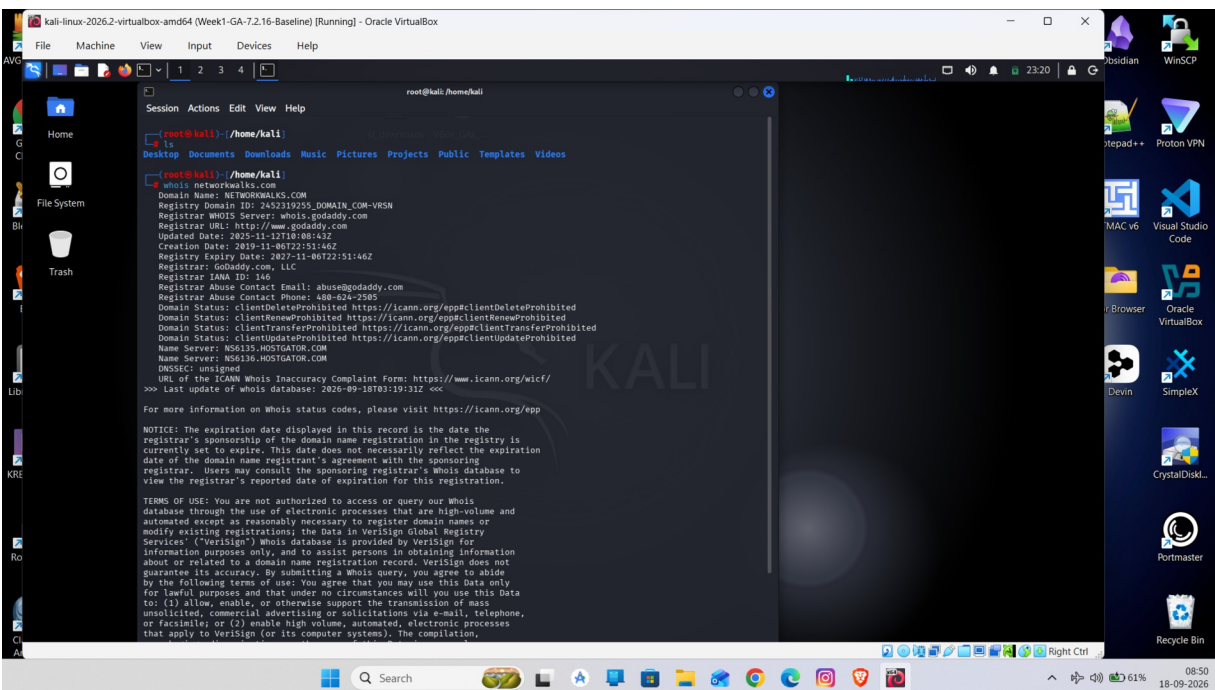
WHY WE PERFORMED THIS: This capture is part of the preserved PM1 evidence chain and supports the registration/delegation observations.

WHAT THIS EVIDENCE SHOWS: Shows public registration, delegation and domain-status information used to establish administrative and infrastructure context.

**Evidence 10 — Additional PM1 evidence**

WHY WE PERFORMED THIS: Supplementary capture supporting the original reconnaissance evidence set and command execution context.

WHAT THIS EVIDENCE SHOWS: Preserves the visual proof required to reproduce, validate and interpret this stage of the assessment.

**5.3 DNS Enumeration and Authoritative Validation**

The local resolver path initially timed out through 10.0.0.1, while explicit queries through 8.8.8.8 resolved the domain to 192.232.216.135. Subsequent authoritative queries to both HostGator nameservers returned the same A record. This is an important methodological lesson: a failed enumeration tool or resolver path does not automatically indicate a target-side DNS failure.

Evidence 05 — DNS record enumeration

WHY WE PERFORMED THIS: The capture supports the A/NS/MX/TXT/SOA record discovery stage.

WHAT THIS EVIDENCE SHOWS: Provides DNS record or authoritative-resolution evidence used to map public infrastructure, mail routing and namespace relationships.

```
(root@kali)~# dig @ns6135.hostgator.com networkwalks.com SOA

; <<>> DiG 9.20.23-1-Debian <<>> @ns6135.hostgator.com networkwalks.com SOA
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 46556
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: f0c2c519941c44bb010000006aacb29ee851712fcbfd8a6b (good)
;; QUESTION SECTION:
;networkwalks.com.                IN      SOA

;; ANSWER SECTION:
networkwalks.com.                86400   IN      SOA      ns6135.hostgator.com. root.gator3068.hostgator.com. 2026091800 86400 7200 3600000 86400

;; Query time: 268 msec
;; SERVER: 50.87.144.87#53(ns6135.hostgator.com) (UDP)
;; WHEN: Thu Sep 17 23:40:14 EDT 2026
;; MSG SIZE rcvd: 157

(root@kali)~# dig networkwalks.com A

;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out

; <<>> DiG 9.20.23-1-Debian <<>> networkwalks.com A
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 61716
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;networkwalks.com.                IN      A

;; ANSWER SECTION:
networkwalks.com.                12894   IN      A          192.232.216.135

;; Query time: 264 msec
;; SERVER: 8.8.8.8#53(8.8.8.8) (UDP)
;; WHEN: Thu Sep 17 23:40:32 EDT 2026
;; MSG SIZE rcvd: 61
```

Evidence 06 — DNS record validation

WHY WE PERFORMED THIS: Additional DNS output used to correlate the public resolver results with target DNS data.

WHAT THIS EVIDENCE SHOWS: Provides DNS record or authoritative-resolution evidence used to map public infrastructure, mail routing and namespace relationships.

```
(root@kali)-[~]
# nslookup networkwalks.com 8.8.8.8
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
Name:   networkwalks.com
Address: 192.232.216.135

(root@kali)-[~]
# nslookup -type=NS networkwalks.com 8.8.8.8
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
networkwalks.com      nameserver = ns6136.hostgator.com.
networkwalks.com      nameserver = ns6135.hostgator.com.

Authoritative answers can be found from:

(root@kali)-[~]
# nslookup -type=MX networkwalks.com 8.8.8.8
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
networkwalks.com      mail exchanger = 0 mail.networkwalks.com.

Authoritative answers can be found from:
```

Evidence 07 — DNS service / record evidence

WHY WE PERFORMED THIS: Supports the broader DNS infrastructure map used in PM1.

WHAT THIS EVIDENCE SHOWS: Provides DNS record or authoritative-resolution evidence used to map public infrastructure, mail routing and namespace relationships.

```
(root@kali)-[~]
# nslookup -type=MX networkwalks.com
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
networkwalks.com      mail exchanger = 0 mail.networkwalks.com.

Authoritative answers can be found from:

(root@kali)-[~]
# nslookup -type=TXT networkwalks.com
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
;; communications error to 10.0.0.1#53: timed out
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
networkwalks.com      text = "v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all"
networkwalks.com      text = "google-site-verification=rr04eRmqHoWY3XemnizDNV4q75X-Ij-mjgEeg-UsYI"

Authoritative answers can be found from:
```

Evidence 13 — DNS validation evidence

WHY WE PERFORMED THIS: Additional capture retained to show the DNS enumeration sequence and returned records.

WHAT THIS EVIDENCE SHOWS: Provides DNS record or authoritative-resolution evidence used to map public infrastructure, mail routing and namespace relationships.

```
(root@kali)~# nslookup -type=TXT networkwalks.com 8.8.8.8
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
networkwalks.com      text = "google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI"
networkwalks.com      text = "v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all"

Authoritative answers can be found from:

(root@kali)~# nslookup -type=SOA networkwalks.com 8.8.8.8
Server:      8.8.8.8
Address:     8.8.8.8#53

Non-authoritative answer:
networkwalks.com
  origin = ns6135.hostgator.com
  mail addr = root.gator3068.hostgator.com
  serial = 2026091800
  refresh = 86400
  retry = 7200
  expire = 3600000
  minimum = 86400

Authoritative answers can be found from:
```

Evidence 11 — dig reference and command context

WHY WE PERFORMED THIS: The capture shows the dig utility context used for DNS-oriented validation and reproducibility.

WHAT THIS EVIDENCE SHOWS: Provides DNS record or authoritative-resolution evidence used to map public infrastructure, mail routing and namespace relationships.

```
Session Actions Edit View Help
DIG(1) BIND 9 DIG(1)

NAME
  dig - DNS lookup utility

SYNOPSIS
  dig [a-server] [-b address] [-c class] [-f filename] [-k filename] [-m] [-p port#] [-q name] [-t type] [-w] [-x addr] [-y [hmac:]name:key] [ [-4] | [-6] ] [name] [type] [class] [query-
  opt ...]

  dig [-h]

  dig [global-queryopt ...] [query ...]

DESCRIPTION
  dig is a flexible tool for interrogating DNS name servers. It performs DNS lookups and displays the answers that are returned from the name server(s) that were queried. Most DNS ad-
  ministrators use dig to troubleshoot DNS problems because of its flexibility, ease of use, and clarity of output. Other lookup tools tend to have less functionality than dig.

  Although dig is normally used with command-line arguments, it also has a batch mode of operation for reading lookup requests from a file. A brief summary of its command-line arguments
  and options is printed when the -h option is given. The BIND 9 implementation of dig allows multiple lookups to be issued from the command line.

  Unless it is told to query a specific name server, dig tries each of the servers listed in /etc/resolv.conf. If no usable server addresses are found, dig sends the query to the local
  host.

  When no command-line arguments or options are given, dig performs an NS query for "." (the root).

  It is possible to set per-user defaults for dig via $(HOME)/.digrc. This file is read and any options in it are applied before the command-line arguments. The -r option disables this
  feature, for scripts that need predictable behavior.

  The IN and CH class names overlap with the IN and CH top-level domain names. Either use the -t and -c options to specify the type and class, use the -q to specify the domain name, or
  use "IN." and "CH." when looking up these top-level domains.
```

Evidence 12 — WHOIS utility reference

WHY WE PERFORMED THIS: This supporting capture documents the command-line environment and utility context used during reconnaissance.

WHAT THIS EVIDENCE SHOWS: Shows public registration, delegation and domain-status information used to establish administrative and infrastructure context.

```

WHOIS(1)
NAME
  whois - client for the whois directory service

SYNOPSIS
  whois [ -h | --host ] HOST [ -p | --port ] PORT [ -a | --addcdhiklmlwrx ] [ -g SOURCE:FIRST-LAST ] [ -i ATTR[,ATTR]... ] [ -s SOURCE[,SOURCE]... ] [ -t TYPE[,TYPE]... ] [ --verbose ] [ --no-recursion ] OBJECT

  whois -q KEYWORD
  whois -t TYPE
  whois -v TYPE
  whois --help
  whois --version

DESCRIPTION
  whois searches for an object in a RFC 3912 database.

  This version of the whois client tries to guess the right server to ask for the specified object. If no guess can be made it will connect to whois.networksolutions.com for NIC handles or whois.arin.net for IPv4 addresses and network names.
  
```

Evidence 14 — DNSRecon utility reference

WHY WE PERFORMED THIS: The capture documents the DNSRecon command-line capability used for automated DNS enumeration.

WHAT THIS EVIDENCE SHOWS: Documents the automated DNS enumeration behavior and resolver/timeout characteristics that were compared with direct DNS validation.

```

DNSRECON(1)
NAME
  dnsrecon - DNS Enumeration and Scanning Tool

SYNOPSIS
  dnsrecon [-h] [-d DOMAIN] [-n NS_SERVER] [-r RANGE] [-D DICTIONARY] [-f] [-a] [-s] [-b] [-k] [-w] [-y] [--threads THREADS] [--lifetime LIFETIME] [--tcp] [--db DB] [-x XML] [-c CSV] [-j JSON] [--lw] [--disable_check_recursion]
  [--disable_check_bindversion] [-v] [-t TYPE]

DESCRIPTION
  dnsrecon is a simple python script that enables to gather DNS-oriented information on a given target.

OPTIONS
  -h, --help
    show help message and exit

  -d DOMAIN, --domain DOMAIN
    Target domain.

  -n NS_SERVER, --name_server NS_SERVER
    Domain server to use. If none is given, the SOA of the target will be used. Multiple servers can be specified using a comma separated list.

  -r RANGE, --range RANGE
    IP range for reverse lookup brute force in formats (first-last) or in (range/bitmask).

  -D DICTIONARY, --dictionary DICTIONARY
    Dictionary file of subdomain and hostnames to use for brute force. Filter out of brute force domain lookup, records that resolve to the wildcard defined IP address when saving records.

  -f
    Filter out of brute force domain lookup, records that resolve to the wildcard defined IP address when saving records.

  -a
    Perform AXFR with standard enumeration.

  -b
    Perform Bing enumeration with standard enumeration.

  -s
    Perform a reverse lookup of IPv4 ranges in the SPF record with standard enumeration.

  -y
    Perform Yandex enumeration with standard enumeration.

  -k
    Perform crt.sh enumeration with standard enumeration.

  -w
    Perform deep whois record analysis and reverse lookup of IP ranges found through Whois when doing a standard enumeration.

  -z
    Performs a DNSSEC zone walk with standard enumeration.

  --threads THREADS
    Number of threads to use in reverse lookups, forward lookups, brute force and SRV record enumeration.

  --lifetime LIFETIME
    Time to wait for a server to respond to a query. default is 3.

  --tcp
    Use TCP protocol to make queries.

  --db DB
    Connect to file to save found records.
  
```

Evidence 15 — DNS resolution validation

WHY WE PERFORMED THIS: Visual proof retained as part of the Week 2 evidence chain.

WHAT THIS EVIDENCE SHOWS: Confirms the target hostname-to-IP relationship and provides the network address used for subsequent authorized enumeration.

```
(root@kali)~# dig @ns6135.hostgator.com networkwalks.com A

; <<>> DiG 9.20.23-1-Debian <<>> @ns6135.hostgator.com networkwalks.com A
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 24681
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 0fbfc9be7a9eecd9010000006aacb264df6dd8abec739cce (good)
;; QUESTION SECTION:
;networkwalks.com.                IN      A

;; ANSWER SECTION:
networkwalks.com.                14400   IN      A      192.232.216.135

;; Query time: 280 msec
;; SERVER: 50.87.144.87#53(ns6135.hostgator.com) (UDP)
;; WHEN: Thu Sep 17 23:39:15 EDT 2026
;; MSG SIZE rcvd: 89

(root@kali)~# dig @ns6136.hostgator.com networkwalks.com A

; <<>> DiG 9.20.23-1-Debian <<>> @ns6136.hostgator.com networkwalks.com A
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 22056
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 9492739e18fff550010000006aacb295aa3b14a23cd9605d (good)
;; QUESTION SECTION:
;networkwalks.com.                IN      A

;; ANSWER SECTION:
networkwalks.com.                14400   IN      A      192.232.216.135

;; Query time: 260 msec
;; SERVER: 192.232.216.131#53(ns6136.hostgator.com) (UDP)
;; WHEN: Thu Sep 17 23:40:05 EDT 2026
;; MSG SIZE rcvd: 89
```

Evidence 17 — DNSRecon execution result

WHY WE PERFORMED THIS: The capture records the DNSRecon enumeration attempt and its timeout behavior.

WHAT THIS EVIDENCE SHOWS: Documents the automated DNS enumeration behavior and resolver/timeout characteristics that were compared with direct DNS validation.

```
(root@kali)~# dnsrecon -d networkwalks.com
2026-09-17T23:36:46.349776-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:36:46.350749-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:36:54.758346-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:36:54.758665-0400 ERROR directly and requests are not being filtered. Increase the timeout from 3.0 second
2026-09-17T23:36:54.758865-0400 ERROR to a higher number with --lifetime <time> option.

(root@kali)~# dnsrecon -d networkwalks.com --lifetime 10 | tee ~/networkwalks_w2pm1_dnsrecon.txt
2026-09-17T23:37:36.889073-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:37:36.889942-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:37:52.312712-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:37:52.313089-0400 ERROR directly and requests are not being filtered. Increase the timeout from 10.0 second
2026-09-17T23:37:52.313220-0400 ERROR to a higher number with --lifetime <time> option.
```

Evidence 18 — DNSRecon explicit-resolver attempt

WHY WE PERFORMED THIS: This capture supports the repeated timeout observation when DNSRecon was pointed at a public resolver.

WHAT THIS EVIDENCE SHOWS: Documents the automated DNS enumeration behavior and resolver/timeout characteristics that were compared with direct DNS validation.

```
(root@kali)-[~]
# dnsrecon -d networkwalks.com -n 8.8.8.8 --lifetime 10
2026-09-17T23:49:31.782870-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:49:31.785223-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:49:47.208823-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:49:47.209095-0400 ERROR directly and requests are not being filtered. Increase the timeout from 10.0 second
2026-09-17T23:49:47.209174-0400 ERROR to a higher number with --lifetime <time> option.

(root@kali)-[~]
# dnsrecon -d networkwalks.com -n 8.8.8.8 --lifetime 10 | tee ~/networkwalks_w2pm1_dnsrecon.txt
2026-09-17T23:49:50.688951-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:49:50.697846-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:50:06.272638-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:50:06.273679-0400 ERROR directly and requests are not being filtered. Increase the timeout from 10.0 second
2026-09-17T23:50:06.273838-0400 ERROR to a higher number with --lifetime <time> option.
```

Evidence 43 — Additional DNSRecon evidence

WHY WE PERFORMED THIS: Supplementary DNSRecon output retained for traceability of the repeated enumeration behavior.

WHAT THIS EVIDENCE SHOWS: Documents the automated DNS enumeration behavior and resolver/timeout characteristics that were compared with direct DNS validation.

```
(root@kali)-[~]
# dnsrecon -d networkwalks.com -n 8.8.8.8 --lifetime 10
2026-09-17T23:49:31.782870-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:49:31.785223-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:49:47.208823-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:49:47.209095-0400 ERROR directly and requests are not being filtered. Increase the timeout from 10.0 second
2026-09-17T23:49:47.209174-0400 ERROR to a higher number with --lifetime <time> option.

(root@kali)-[~]
# dnsrecon -d networkwalks.com -n 8.8.8.8 --lifetime 10 | tee ~/networkwalks_w2pm1_dnsrecon.txt
2026-09-17T23:49:50.688951-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:49:50.697846-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:50:06.272638-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:50:06.273679-0400 ERROR directly and requests are not being filtered. Increase the timeout from 10.0 second
2026-09-17T23:50:06.273838-0400 ERROR to a higher number with --lifetime <time> option.
```

Evidence 50 — Repeated DNSRecon evidence

WHY WE PERFORMED THIS: Additional capture supporting the DNSRecon timeout comparison and evidence preservation.

WHAT THIS EVIDENCE SHOWS: Documents the automated DNS enumeration behavior and resolver/timeout characteristics that were compared with direct DNS validation.

```
(root@kali)~# dnsrecon -d networkwalks.com
2026-09-17T23:36:46.349776-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:36:46.350749-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:36:54.758346-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:36:54.758665-0400 ERROR directly and requests are not being filtered. Increase the timeout from 3.0 second
2026-09-17T23:36:54.758865-0400 ERROR to a higher number with --lifetime <time> option.

(root@kali)~# dnsrecon -d networkwalks.com --lifetime 10 | tee ~/networkwalks_w2pm1_dnsrecon.txt
2026-09-17T23:37:36.889073-0400 INFO Starting enumeration for domain: networkwalks.com
2026-09-17T23:37:36.889942-0400 INFO std: Performing General Enumeration against: networkwalks.com...
2026-09-17T23:37:52.312712-0400 ERROR A timeout error occurred please make sure you can reach the target DNS Servers
2026-09-17T23:37:52.313089-0400 ERROR directly and requests are not being filtered. Increase the timeout from 10.0 second
2026-09-17T23:37:52.313220-0400 ERROR to a higher number with --lifetime <time> option.
```

5.4 Web Technology Fingerprinting

WhatWeb identified an externally fingerprintable web stack including Apache and WordPress-related technology signals. The evidence also indicated WordPress Download Manager behavior. A fingerprint is useful for narrowing later authorized security research, but the fingerprint alone is not proof of a vulnerable component.

Evidence 03 — WhatWeb fingerprint

WHY WE PERFORMED THIS: This is the primary visual evidence for the externally observable web technology stack.

WHAT THIS EVIDENCE SHOWS: Identifies externally observable web technologies and application fingerprints that can guide authorized patch and configuration review.

```
(root@kali)~# whatweb https://networkwalks.com
https://networkwalks.com [200 OK] Apache, Bootstrap[7.1.1], Cookies[_wpdm_client], Country[UNITED STATES], Email[info@networkwalks.com], Frame, Google-Tag-Manager, HTML5, HTTPServer[Apache], HttpOnly[_wpdm_client], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.1.1,WordPress Download Manager 3.3.58], Open-Graph-Protocol[website], Script[668NR-IPID&pid=Var2=58511&pid=prtVar2=7&pid=scvVar2=12,application/json,application/ld+json,module,speculat
ionrules,text/javascript], Title[Networkwalks Academy], UncommonHeaders[permissions-policy,link,upgrade,referrer-policy,x-endurance-cache-level,x-nginx-cache], WordPress[7.1.1]
```

5.5 HTTP Metadata, REST API References and Cookies

The cURL HEAD response returned HTTP/2 200 and exposed server/header metadata. The Link header referenced the WordPress REST API, including /wp-json/ and a page-related route. The __wpdm_client cookie was observed with Secure and HttpOnly attributes. These observations identify interfaces and metadata an attacker would investigate next, but they do not demonstrate unauthorized access.

Evidence 16 — cURL HTTP header inspection

WHY WE PERFORMED THIS: Primary evidence for the HTTP response, Link header, cookie and server metadata.

WHAT THIS EVIDENCE SHOWS: Captures the actual HTTP response metadata, headers, cookies and implementation signals returned by the target.

```
(root@kali)~# curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel="shortlink"
set-cookie: _wpdm_client=43clab7bebcfc16d229d405abf6062b; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 18 Sep 2026 03:59:54 GMT
server: Apache
```

Evidence 45 — Additional HTTP header evidence

WHY WE PERFORMED THIS: Supplementary cURL capture retained to corroborate the HTTP metadata observations.

WHAT THIS EVIDENCE SHOWS: Captures the actual HTTP response metadata, headers, cookies and implementation signals returned by the target.

```
(root@kali)~# curl -I https://networkwalks.com
HTTP/2 200
permissions-policy: private-state-token-redemption=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com"), private-state-token-issuance=(self "https://www.google.com" "https://www.gstatic.com" "https://recaptcha.net" "https://challenges.cloudflare.com" "https://hcaptcha.com")
link: <https://networkwalks.com/wp-json/>; rel="https://api.w.org/", <https://networkwalks.com/wp-json/wp/v2/pages/53>; rel="alternate"; title="JSON"; type="application/json", <https://networkwalks.com/>; rel="shortlink"
set-cookie: _wpdm_client=43clab7bebcfc16d229d405abf6062b; path=/; domain=networkwalks.com; secure; HttpOnly
referrer-policy: no-referrer-when-downgrade
x-endurance-cache-level: 0
x-nginx-cache: WordPress
content-type: text/html; charset=UTF-8
date: Fri, 18 Sep 2026 03:59:54 GMT
server: Apache
```

Evidence 08 — Supplementary PM1 capture

WHY WE PERFORMED THIS: Additional preserved PM1 screenshot retained as part of the reconnaissance evidence chain.

WHAT THIS EVIDENCE SHOWS: Preserves the visual proof required to reproduce, validate and interpret this stage of the assessment.

```
NSLOOKUP(1)                                BIND 9                                NSLOOKUP(1)

NAME
    nslookup - query Internet name servers interactively

SYNOPSIS
    nslookup [-option] [name | -] [server]

DESCRIPTION
    nslookup is a program to query Internet domain name servers. nslookup has two modes: interactive and non-interactive. Interactive mode allows the user to query name servers for information about various hosts and domains or to print a list of hosts in a domain. Non-interactive mode prints just the name and requested information for a host or domain.

ARGUMENTS
    Interactive mode is entered in the following cases:
    a. when no arguments are given (the default name server is used);
    b. when the first argument is a hyphen (-) and the second argument is the host name or Internet address of a name server.

    Non-interactive mode is used when the name or Internet address of the host to be looked up is given as the first argument. The optional second argument specifies the host name or address of a name server.

    Options can also be specified on the command line if they precede the arguments and are prefixed with a hyphen. For example, to change the default query type to host information, with an initial timeout of 10 seconds, type:
        nslookup -query=hhinfo -timeout=10

    The -version option causes nslookup to print the version number and immediately exit.

INTERACTIVE COMMANDS
    host [server]
        This command looks up information for host <std-iscan-host> using the current default server or using server, if specified. If host <std-iscan-host> is an Internet address and the query type is A or PTR, the name of the host is returned. If host <std-iscan-host> is a name and does not have a trailing period (.), the search list is used to qualify the name.

        To look up a host not in the current domain, append a period to the name.

    server domain | lserver domain
        These commands change the default server to domain; lserver uses the initial server to look up information about domain, while server uses the current default server. If an authoritative answer cannot be found, the names of servers that might have the answer are returned.
```

Evidence 20 — theHarvester / reconnaissance environment evidence

WHY WE PERFORMED THIS: Supporting capture connecting the broader reconnaissance tooling environment to the evidence set.

WHAT THIS EVIDENCE SHOWS: Preserves the passive-reconnaissance tooling context used to support repeatable public-intelligence collection and correlation.

```
(root@kali)~# theHarvester -h
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester *
* theHarvester 4.10.1 *
* Coded by Christian Martorella *
* Edge-Security Research *
* cmartorella@edge-security.com *
*****
usage: theHarvester [-h] [-d DOMAIN [-l LIMIT]] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]

theHarvester is used to gather open source intelligence (OSINT) on a company or domain.

options:
  -h, --help            show this help message and exit
  -d, --domain DOMAIN    Company name or domain to search.
  -l, --limit LIMIT      Limit the number of search results, default=500.
  -s, --start START      Start with result number X, default=0.
  -p, --proxies           Use proxies for requests, enter proxies in proxies.yaml.
  -s, --shodan           Use Shodan to query discovered hosts.
  --screenshot SCREENSHOT
                        Take screenshots of resolved domains specify output directory: --screenshot output_directory
  -e, --dns-server DNS_SERVER
                        DNS server to use for lookup.
  -t, --take-over        Check for takeovers.
  -r, --dns-resolve [DNS_RESOLVE]
                        Perform DNS resolution on subdomains with a resolver list or passed in resolvers, default False.
  -n, --dns-lookup       Enable DNS server lookup, default False.
  -c, --dns-brute        Perform a DNS brute force on the domain.
  -f, --filename FILENAME
                        Save the results to an XML and JSON file.
  -w, --wordlist WORDLIST
                        Specify a wordlist for API endpoint scanning.
  -a, --api-scan         Scan for API endpoints.
  -q, --quiet            Suppress missing API key warnings and reading the api-keys file.
  -b, --source SOURCE    Have been pwned, hudsonrock, hunter, hunterhow, intelx, leakix, leaklookup, netlas, onyphe, otx, pentesttools, projectdiscovery, rapiddns, robtex, rocketreach, securityscorecard, security
```

5.6 WAF Detection

WAFW00F identified ModSecurity (SpiderLabs). This is a defensive-architecture observation. The capture does not establish WAF version, rule coverage, logging quality, bypass resistance or the security of the application behind it. **Evidence 02 — WAFW00F result**

WHY WE PERFORMED THIS: Primary evidence for the ModSecurity (SpiderLabs) identification.

WHAT THIS EVIDENCE SHOWS: Records the defensive technology fingerprint observed at the web boundary; it does not by itself prove security effectiveness.

```
(root@kali)~# wafw00f https://networkwalks.com

{ WOOF! }

404 Hack Not Found
405 Not Allowed
403 Forbidden
502 Bad Gateway
500 Internal Error

~ WAFW00F : v2.4.2 ~
The Web Application Firewall Fingerprinting Toolkit

[*] Checking https://networkwalks.com
[+] The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.
[~] Number of requests: 2
```

5.7 SpiderFoot — Passive Reconnaissance Support

SpiderFoot 4.0.0 was available at `/usr/bin/spiderfoot`, with its main script located under `/usr/share/spiderfoot/sf.py`. The captured PM1 work used SpiderFoot for capability discovery and reconnaissance preparation: version, executable location, help, modules and event/data types were reviewed. This report uses that evidence as tooling context and does not fabricate a SpiderFoot target-scan dataset that is not present in the captured evidence.

5.8 PM1 Analyst Interpretation

From an attacker perspective, PM1 reduces uncertainty. WHOIS provides administrative and provider context; DNS maps public infrastructure; web fingerprinting identifies technology; cURL exposes application metadata and interface references; WAFW00F reveals a defensive layer; and SpiderFoot provides an OSINT workflow. The security value comes from correlation across these signals, not from any one output.

6. W2-PM2 — GHDB-Based Footprinting Attacks

PM2 focuses on search-engine footprinting using the Google Hacking Database (GHDB). The supplied Networkwalks practical defines two exercises: locating indexed exposed security-camera interfaces and locating indexed mathematics PDF listings. The important technical lesson is that search-engine indexing can reveal resources without directly probing the target infrastructure.

6.1 How the Technique Works



The report deliberately treats search results as candidate intelligence rather than automatically as live vulnerable assets. An indexed page can be stale, misattributed, inaccessible, or outside the authorized target. That distinction is essential for professional reconnaissance reporting.

6.2 Task 1 — Indexed Exposed-Camera Interfaces

The supplied practical asks the learner to use GHDB search operators to identify live exposed camera links and record the corresponding search pattern. In an authorized assessment, the correct workflow is to document the search methodology, verify that a result is actually reachable, establish that it belongs to the authorized scope, and only then classify the security significance.

6.3 Task 2 — Indexed Mathematics PDF Listings

The second exercise applies search-engine indexing to downloadable mathematics PDF listings. From a defensive perspective, the same technique can identify accidentally indexed directories, documents and file repositories that organizations may not intend to expose.

6.4 PM2 Outcome

PM2 demonstrates the reconnaissance principle that the public search index itself can become an intelligence source. The useful outcome is the ability to move from a search result to an evidence-backed exposure assessment while maintaining scope and attribution discipline.

Evidence 44 — Week 2 project-module reference

WHY WE PERFORMED THIS: This capture documents the Week 2 module structure used to frame PM2 and the surrounding assessment workflow.

WHAT THIS EVIDENCE SHOWS: Preserves the supplied Week 2 module context used to map the practical activities to the assessment workflow.

You are viewing 01. Waqas Karim CCIE's screen REC View Options

Week 2 - Project Modules

Choose any 1 module from electives + both essential projects (total 3 modules minimum).

Electives (choose at least 1):

- W2-PM1: Footprinting with multiple Kali tools (6 tools on networkwalks.com)
- W2-PM2: GHDB based Footprinting Attacks
- W2-PM3: Maltego based Footprinting Attacks
- W2-PM4: theHarvester based Footprinting Attacks

Essentials (both required):

- W2-PM5: Zenmap based Network Scanning
- W2-PM-FINAL: Write a detailed report covering the modules you chose

***Note:**

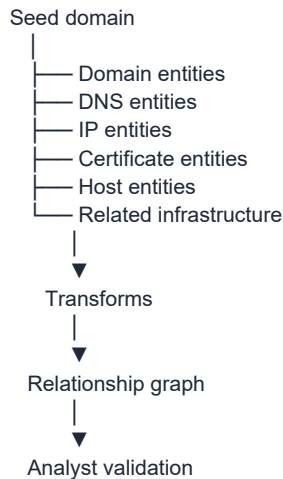
- Report should be based only on the modules you elect and complete.
- You can choose more than one module from Electives, but a minimum of one is required.
- Both Essential modules must be completed (no choice).

www.networkwalks.com

7. W2-PM3 — Maltego-Based Footprinting Attacks

PM3 is the relationship-analysis component of the Week 2 reconnaissance workflow. Maltego is valuable when the analyst needs to connect entities rather than inspect one artifact at a time.

7.1 Entity and Relationship Model



For this Week 2 report, the PM3 analytical model is integrated with the evidence from PM1 and PM4: DNS records, certificate-derived names, provider indicators and hostnames can be represented as related entities, then independently validated before being treated as an assessment finding.

7.2 Why Maltego Matters

Analyst question	Maltego-style treatment	Validation required
Which assets appear related?	Connect domains, hosts, certificates and IPs	Independent DNS/CT/network evidence
Is a hostname current?	Represent it as an entity	Current DNS/addressability
Is an infrastructure relationship meaningful?	Trace shared provider/certificate relationships	Ownership or authoritative evidence
Does a relationship imply a vulnerability?	No — relationship is only a pivot	Application/configuration validation

7.3 PM3 Outcome

The practical outcome of PM3 in the consolidated report is a relationship-oriented way of organizing reconnaissance data. The report does not convert an inferred relationship into a confirmed ownership or vulnerability claim without independent validation.

8. W2-PM4 — Passive Reconnaissance & Certificate Intelligence

PM4 expanded the footprint without relying primarily on direct target interaction. Certificate Transparency and passive discovery sources were used to recover hostnames and historical namespace information.

8.1 PM4 Tool Baseline

The captured environment recorded theHarvester 4.10.1, Python 3.13.12 and Subfinder 2.16.0. These tools provide different passive perspectives and therefore work well as correlation sources.

Evidence 19 — theHarvester baseline

WHY WE PERFORMED THIS: Primary evidence for the theHarvester tool environment and passive-source workflow.

WHAT THIS EVIDENCE SHOWS: Establishes the passive-collection environment and supports reproducibility of the hostname, email and certificate-source workflow.

```
(root@kali)-[~]
# theHarvester -d networkwalks.com -b certspotter -l 100
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*                                     *
* [T]h[e]H[ar]v[e]s[t]e[r]          *
* [T]h[e]H[ar]v[e]s[t]e[r]          *
* [T]h[e]H[ar]v[e]s[t]e[r]          *
* [T]h[e]H[ar]v[e]s[t]e[r]          *
* theHarvester 4.10.1                *
* Coded by Christian Martorella      *
* Edge-Security Research             *
* cmartorella@edge-security.com      *
*                                   *
*****

[*] Target: networkwalks.com

    Searching results.
[*] Searching Certspotter.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] Hosts found: 1
_____.
*.networkwalks.com
```

Evidence 23 — Passive reconnaissance tool context

WHY WE PERFORMED THIS: Supporting capture for theHarvester and certificate-source collection.

WHAT THIS EVIDENCE SHOWS: Preserves the visual proof required to reproduce, validate and interpret this stage of the assessment.

```
(root@kali)-[~]
# which theHarvester
/usr/bin/theHarvester

(root@kali)-[~]
# python3 --version
Python 3.13.12
```

Evidence 24 — Certificate-source workflow context

WHY WE PERFORMED THIS: Additional PM4 evidence retained for reproducibility.

WHAT THIS EVIDENCE SHOWS: Preserves the passive-reconnaissance workflow and source context used to make the namespace analysis reproducible.

```
(root@kali) ~# theHarvester --version
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester 4.18.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****
usage: theHarvester [-h] -d DOMAIN [-l LIMIT] [-s START] [-p] [-s] [--screenshot SCREENSHOT] [-e DNS_SERVER] [-t] [-r [DNS_RESOLVE]] [-n] [-c] [-f FILENAME] [-w WORDLIST] [-a] [-q] [-b SOURCE]
theHarvester: error: the following arguments are required: -d/--domain
```

Evidence 48 — Subfinder baseline

WHY WE PERFORMED THIS: Primary evidence for the Subfinder installation and command-line environment.

WHAT THIS EVIDENCE SHOWS: Records passive subdomain discovery and provides an independent source for cross-checking certificate-derived hostnames.

```
(root@kali) ~# subfinder -version
[INF] Current Version: v2.16.0
[INF] Subfinder Config Directory: /home/kali/.config/subfinder

(root@kali) ~# which subfinder
/usr/bin/subfinder
```

8.2 CertSpotter

The CertSpotter source returned the wildcard namespace *.networkwalks.com. A wildcard certificate is useful as namespace intelligence, but it does not prove that every possible hostname under the wildcard is currently active.

8.3 crt.sh Adapter and Direct Certificate Transparency

The theHarvester crt.sh adapter returned no hosts in the recorded invocation. The result was treated as an adapter-level outcome rather than as evidence that the global Certificate Transparency dataset was empty. A direct crt.sh query subsequently produced certificate records that could be normalized and analyzed.

Evidence 22 — crt.sh adapter result

WHY WE PERFORMED THIS: Shows the adapter-level result that was compared against direct Certificate Transparency collection.

WHAT THIS EVIDENCE SHOWS: Provides certificate-derived hostnames and temporal context used to separate current namespace evidence from historical intelligence.

```
(root@kali)-[~]
# theHarvester -d networkwalks.com -b crtsh -l 100
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*
* [TheHarvester]
* [TheHarvester]
* [TheHarvester]
* [TheHarvester]
* [TheHarvester]
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: networkwalks.com

[*] Searching CRTsh.

[*] No IPs found.

[*] No emails found.

[*] No people found.

[*] No hosts found.
```

Evidence 21 — Direct Certificate Transparency collection

WHY WE PERFORMED THIS: Primary evidence for the direct CT dataset.

WHAT THIS EVIDENCE SHOWS: Provides certificate-derived hostnames and temporal context used to separate current namespace evidence from historical intelligence.

```

2025-11-17T18:05:14 2026-09-28T18:05:14 CnUS, ST=Arizona, L=Scottsdale, O=GoDaddy.com, Inc., OU=http://certs.godaddy.com/repository/, Cn=Go Daddy Secure Certificate Authority - G2 www.networkwalks.com www.networkwalks.com
2025-12-21T02:17:22 2026-09-21T02:17:22 CnUS, O=Let's Encrypt, CN=R3 networkwalks.com *.networkwalks.com, networkwalks.com
2026-02-28T02:17:22 2026-09-21T02:17:22 CnUS, O=Let's Encrypt, CN=R3 networkwalks.com *.networkwalks.com, networkwalks.com
2026-04-23T01:21:19 2026-07-22T01:21:18 CnUS, O=Let's Encrypt, CN=R3 *.networkwalks.com *.networkwalks.com, networkwalks.com
2026-06-23T01:07:20 2026-09-21T01:07:20 CnUS, O=Let's Encrypt, CN=R2 networkwalks.com *.networkwalks.com, networkwalks.com
2026-08-23T01:09:24 2026-11-21T01:09:23 CnUS, O=Let's Encrypt, CN=R2 *.networkwalks.com *.networkwalks.com, networkwalks.com

PMA-006: Certificate Timeline without Issuer
2019-11-08T04:35:05 2026-02-06T04:35:05 networkwalks.aurume.com autodiscover.networkwalks.com, cpanel.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.networkwalks.com, webmail.networkwalks.com, www.net
2020-01-08T04:47:14 2026-04-07T04:47:14 networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.networkwalks.com, webmail.networkwalks.com, www.netwo
2020-02-04T04:08:00 2021-02-03T2:59:59 networkwalks.com networkwalks.com, www.networkwalks.com
2020-03-09T04:32:40 2026-06-07T04:32:40 networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.networkwalks.com, webmail.networkwalks.com, www.netwo
2020-05-18T05:38:00 2026-06-08T05:38:00 autodiscover.networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.networkwalks.com, mail.networkwalks.com, networkwalks.com,
networkwalks.com,
2020-07-11T11:07:16 2026-10-09T11:07:16 networkwalks.aurume.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.networkwalks.com, mail.networkwalks.com, networkwalks.com,
networkwalks.com,
2020-09-11T08:57:28 2026-12-10T08:57:28 webdisk.networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.networkwalks.com, mail.networkwalks.com, networkwalks.com,
networkwalks.com,
2020-11-12T04:36:30 2021-02-10T04:36:30 networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.
2021-01-12T06:09:45 2021-04-12T06:09:45 mail.networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.
2021-03-15T02:24:52 2021-06-13T02:24:52 mail.aurume.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.
2021-05-16T04:22:48 2021-08-14T04:22:48 atechacademy.com *.networkwalks.com, networkwalks.com
2021-07-16T04:21:17 2021-10-14T04:21:15 *.aurume.com *.networkwalks.com, networkwalks.com
2021-09-15T04:08:55 2021-12-14T04:08:54 *.aurume.com *.networkwalks.com, networkwalks.com
2021-11-15T04:44:41 2022-02-13T04:44:40 atechacademy.com *.networkwalks.com, networkwalks.com
2022-01-15T04:38:45 2022-04-15T04:38:44 *.aurume.com *.networkwalks.com, networkwalks.com
2022-03-17T04:02:41 2022-06-15T04:02:40 *.aurume.com *.networkwalks.com, networkwalks.com
2022-05-17T04:01:03 2022-08-15T04:01:02 atechacademy.com *.networkwalks.com, networkwalks.com
2022-07-17T03:30:42 2022-10-15T03:30:41 atechacademy.com *.networkwalks.com, networkwalks.com
2022-09-17T03:31:53 2022-12-16T03:31:52 *.aurume.com *.networkwalks.com, networkwalks.com
2022-11-18T03:22:58 2023-02-16T03:22:57 *.aurume.com *.networkwalks.com, networkwalks.com
2023-01-18T03:11:42 2023-04-18T03:11:41 *.aurume.com *.networkwalks.com, networkwalks.com
2023-03-18T03:58:13 2023-06-18T03:58:12 *.atechacademy.com *.networkwalks.com, networkwalks.com
2023-05-21T01:48:37 2023-08-19T01:48:36 *.aurume.com *.networkwalks.com, networkwalks.com
2023-07-21T01:44:01 2023-10-19T01:44:00 networkwalks.com *.networkwalks.com, networkwalks.com
2023-09-20T01:58:57 2023-12-19T01:58:56 networkwalks.com *.networkwalks.com, networkwalks.com
2023-11-20T03:26:26 2024-02-18T03:26:25 *.networkwalks.com *.networkwalks.com, networkwalks.com
2024-01-20T03:28:33 2024-04-19T03:28:32 *.aurume.com *.networkwalks.com, networkwalks.com
2024-03-22T02:14:03 2024-06-20T02:14:02 *.aurume.com *.networkwalks.com, networkwalks.com
2024-05-22T02:29:42 2024-08-20T02:29:41 www.networkwalks.aurume.com *.networkwalks.com, networkwalks.com
2024-07-22T02:29:42 2024-10-20T02:29:41 networkwalks.com *.networkwalks.com, networkwalks.com
2024-09-21T02:52:22 2024-12-20T02:52:21 *.aurume.com *.networkwalks.com, networkwalks.com
2024-11-21T02:42:59 2025-02-19T02:42:58 *.aurume.com *.networkwalks.com, networkwalks.com
2025-01-21T03:08:25 2025-04-21T03:08:24 www.zig19.aurume.com *.networkwalks.com, networkwalks.com
2025-03-23T02:11:42 2025-06-21T02:11:41 *.aurume.com *.networkwalks.com, networkwalks.com
2025-05-23T02:18:04 2025-08-21T02:18:03 *.aurume.com *.networkwalks.com, networkwalks.com
2025-07-23T02:07:30 2025-10-21T02:07:29 *.networkwalks.com *.networkwalks.com, networkwalks.com
2025-09-23T02:04:52 2025-12-21T02:04:51 networkwalks.com *.networkwalks.com, networkwalks.com
2025-11-11T18:05:14 2026-03-21T02:17:22 www.networkwalks.com www.networkwalks.com
2025-12-21T02:17:22 2026-05-21T02:17:21 networkwalks.com *.networkwalks.com, networkwalks.com
2026-02-21T02:17:22 2026-05-21T02:17:21 networkwalks.com *.networkwalks.com, networkwalks.com

```

Evidence 25 — Certificate names / historical data

WHY WE PERFORMED THIS: Supports hostname extraction and historical certificate analysis.

WHAT THIS EVIDENCE SHOWS: Provides certificate-derived hostnames and temporal context used to separate current namespace evidence from historical intelligence.

```

root@kali: ~
└─ nano CURL_PMA.sh
└─ /CURL_PMA.sh
root@kali: ~
└─ /CURL_PMA.sh
PMA-006: Certificate Timeline with Issuer
2019-11-08T04:35:05 2026-02-06T04:35:05 CnUS, O=Let's Encrypt, CN=Let's Encrypt Authority X3 networkwalks.aurume.com autodiscover.networkwalks.com, cpanel.networkwalks.com, mail.networkwalks.com, networkwalks.com, web
2020-01-08T04:47:14 2026-04-07T04:47:14 workwalks.com, www.networkwalks.com networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.net
2020-02-04T04:08:00 2021-02-03T2:59:59 CnGB, ST=Greater Manchester, L=Salford, O=Secure RSA Domain Validation Secure Server CA networkwalks.com networkwalks.com, www.networkwalks.com,
s.com, www.networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.networkwalks.com, mail.networkwalks.com, networkwalks.com, webdisk.net
2020-03-09T04:32:40 2026-06-07T04:32:40 CnUS, O=Let's Encrypt, CN=Let's Encrypt Authority X3 networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.net
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2020-09-11T08:57:28 2026-12-10T08:57:28 on, networkwalks.com, webdisk.networkwalks.com CnUS, O=Let's Encrypt, CN=Let's Encrypt Authority X3 webmail.networkwalks.com, www.networkwalks.com autodiscover.networkwalks.com, cpanel.networkwalks.com, cpanelendars.networkwalks.com, cpccontacts.net
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2021-05-16T04:22:48 2021-08-14T04:22:48 CnUS, O=Let's Encrypt, CN=R3 atechacademy.com *.networkwalks.com, networkwalks.com
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2021-09-15T04:08:55 2021-12-14T04:08:54 CnUS, O=Let's Encrypt, CN=R2 *.aurume.com *.networkwalks.com, networkwalks.com
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2022-09-17T03:31:53 2022-12-16T03:31:52 CnUS, O=Let's Encrypt, CN=R3 *.aurume.com *.networkwalks.com, networkwalks.com
2022-11-18T03:22:58 2023-02-16T03:22:57 CnUS, O=Let's Encrypt, CN=R3 *.aurume.com *.networkwalks.com, networkwalks.com
2023-01-18T03:11:42 2023-04-18T03:11:41 CnUS, O=Let's Encrypt, CN=R3 *.aurume.com *.networkwalks.com, networkwalks.com
2023-03-18T03:58:13 2023-06-18T03:58:12 CnUS, O=Let's Encrypt, CN=R3 *.atechacademy.com *.networkwalks.com, networkwalks.com
2023-05-21T01:48:37 2023-08-19T01:48:36 CnUS, O=Let's Encrypt, CN=R3 *.aurume.com *.networkwalks.com, networkwalks.com
2023-07-21T01:44:01 2023-10-19T01:44:00 CnUS, O=Let's Encrypt, CN=R3 networkwalks.com *.networkwalks.com, networkwalks.com
2023-09-20T01:58:57 2023-12-19T01:58:56 CnUS, O=Let's Encrypt, CN=R3 networkwalks.com *.networkwalks.com, networkwalks.com
2023-11-20T03:26:26 2024-02-18T03:26:25 CnUS, O=Let's Encrypt, CN=R3 *.networkwalks.com *.networkwalks.com, networkwalks.com
2024-01-20T03:28:33 2024-04-19T03:28:32 CnUS, O=Let's Encrypt, CN=R3 *.aurume.com *.networkwalks.com, networkwalks.com
2024-03-22T02:14:03 2024-06-20T02:14:02 CnUS, O=Let's Encrypt, CN=R3 *.aurume.com *.networkwalks.com, networkwalks.com
2024-05-22T02:29:42 2024-08-20T02:29:41 CnUS, O=Let's Encrypt, CN=R3 www.networkwalks.aurume.com *.networkwalks.com, networkwalks.com
2024-07-22T02:29:42 2024-10-20T02:29:41 CnUS, O=Let's Encrypt, CN=R11 www.networkwalks.aurume.com *.networkwalks.com, networkwalks.com
2024-09-21T02:52:22 2024-12-20T02:52:21 CnUS, O=Let's Encrypt, CN=R10 *.aurume.com *.networkwalks.com, networkwalks.com
2024-11-21T02:42:59 2025-02-19T02:42:58 CnUS, O=Let's Encrypt, CN=R10 *.aurume.com *.networkwalks.com, networkwalks.com
2025-01-21T03:08:25 2025-04-21T03:08:24 CnUS, O=Let's Encrypt, CN=R10 www.zig19.aurume.com *.networkwalks.com, networkwalks.com
2025-03-23T02:11:42 2025-06-21T02:11:41 CnUS, O=Let's Encrypt, CN=R11 *.aurume.com *.networkwalks.com, networkwalks.com
2025-05-23T02:18:04 2025-08-21T02:18:03 CnUS, O=Let's Encrypt, CN=R10 *.networkwalks.com *.networkwalks.com, networkwalks.com
2025-07-23T02:07:30 2025-10-21T02:07:29 CnUS, O=Let's Encrypt, CN=R11 *.aurume.com *.networkwalks.com, networkwalks.com
2025-09-23T02:04:52 2025-12-21T02:04:51 CnUS, O=Let's Encrypt, CN=R10 *.networkwalks.com *.networkwalks.com, networkwalks.com
2025-11-11T18:05:14 2026-03-21T02:17:22 CnUS, O=Let's Encrypt, CN=R11 *.aurume.com *.networkwalks.com, networkwalks.com
2025-12-21T02:17:22 2026-05-21T02:17:21 CnUS, O=Let's Encrypt, CN=R11 *.aurume.com *.networkwalks.com, networkwalks.com
2026-02-21T02:17:22 2026-05-21T02:17:21 CnUS, O=Let's Encrypt, CN=R11 *.aurume.com *.networkwalks.com, networkwalks.com

```

Evidence 26 — Certificate timeline evidence

WHY WE PERFORMED THIS: Additional CT evidence used for temporal and namespace interpretation.

WHAT THIS EVIDENCE SHOWS: Provides certificate-derived hostnames and temporal context used to separate current namespace evidence from historical intelligence.

```

root@kali: ~
Session Actions Edit View Help
curl -s -o /tmp/networkwalks.crt.json https://crt.sh/?q=%25.networkwalks.com&output=json
jq -r '{.not_before, .not_after, .issuer_name, .common_name, .name_value | gsub(" ", "")}' /tmp/networkwalks.crt.json | sort -u
jq -r '{.not_before, .not_after, .common_name, .name_value | gsub(" ", "")}' /tmp/networkwalks.crt.json | sort -u

```

8.4 Normalized Certificate-Derived Namespace

```

networkwalks.com
├── *.networkwalks.com
├── www.networkwalks.com
├── mail.networkwalks.com
├── cpanel.networkwalks.com
├── webmail.networkwalks.com
├── webdisk.networkwalks.com
├── autodiscover.networkwalks.com
├── cpcontacts.networkwalks.com
└── cpcalendars.networkwalks.com

```

Ten unique certificate-derived names were normalized from the direct CT data. The service-oriented pattern is notable because it maps common web, mail and hosting-management namespaces.

8.5 Historical Certificate Intelligence

Historical CT records showed recurring service-oriented names, particularly in earlier certificate periods. Historical evidence is useful for reconstructing infrastructure evolution, but it must not be presented as proof that every historical hostname remains live.

8.6 Subfinder Cross-Source Correlation

Subfinder v2.16.0 returned eight hostnames. Eight of the ten normalized CT names were independently observed by Subfinder, producing the strongest cross-source overlap in PM4.

Evidence 46 — Subfinder passive discovery

WHY WE PERFORMED THIS: Primary evidence for the eight-name Subfinder result.

WHAT THIS EVIDENCE SHOWS: Records passive subdomain discovery and provides an independent source for cross-checking certificate-derived hostnames.

```
(root@kali)-[~]
# subfinder -d networkwalks.com -all -silent
cpcalendars.networkwalks.com
www.networkwalks.com
webdisk.networkwalks.com
autodiscover.networkwalks.com
cpanel.networkwalks.com
mail.networkwalks.com
webmail.networkwalks.com
cpcontacts.networkwalks.com
```

Evidence 51 — Subfinder methodology / output

WHY WE PERFORMED THIS: Additional capture supporting the passive discovery and reproducibility of the PM4 result.

WHAT THIS EVIDENCE SHOWS: Records passive subdomain discovery and provides an independent source for cross-checking certificate-derived hostnames.

```
(root@kali)-[~]
# subfinder -h
Subfinder is a subdomain discovery tool that discovers subdomains for websites by using passive online sources.

Usage:
  subfinder [flags]

Flags:
  -h, --help            help for subfinder
  -i, --input string     domains to find subdomains for
  -d, --list string      file containing list of domains for subdomain discovery

SOURCE:
  -s, --sources string[] specific sources to use for discovery (-s crtsh,github). Use -ls to display all available sources.
  -r, --recursive         use only sources that can handle subdomains recursively rather than both recursive and non-recursive sources
  -a, --all              use all sources for enumeration (slow)
  -e, --exclude-sources string[] sources to exclude from enumeration (-es alienvault,zoomeyeapi)

FILTER:
  -m, --match string[]  subdomain or list of subdomain to match (file or comma separated)
  -f, --filter string[] subdomain or list of subdomain to filter (file or comma separated)

RATE-LIMIT:
  -rl, --rate-limit int maximum number of http requests to send per second (global)
  -rls, --rate-limits value maximum number of http requests to send per second for providers in key-value format (--rls hackertarget=10/m) (default ["github=30/m", "fullhunt=60/m", "pugrecon=10/s", "robtex=10446744073709551615/ms", "sec
  -t int number of concurrent goroutines for resolving (-active only) (default 10)

UPDATE:
  -up, --update          update subfinder to latest version
  -dc, --disable-update-check disable automatic subfinder update check

OUTPUT:
  -o, --output string    file to write output to
  -d, --json             write output in JSON(lines) format
  -dd, --output-dir string directory to write output (-dd only)
  -cs, --collect-sources include all sources in the output (-json only)
  -li, --ip             include host IP in output (-active only)

CONFIGURATION:
  -config string          flag config file (default "/home/kali/.config/subfinder/config.yaml")
  -pc, --provider-config string provider config file (default "/home/kali/.config/subfinder/provider-config.yaml")
  -r string[]            comma separated list of resolvers to use
  -rl string[]          file containing list of resolvers to use
  -ma, --active          display active subdomains only
  -proxy string          http proxy to use with subfinder
  -ei, --exclude-ip      exclude IP from the list of domains
  -mr, --max-results int limit the number of results per source (0 = unlimited; honored by paginating sources)

DEBUG:
  -silent              show only subdomains in output
  -v                  show version of subfinder
  -v, --verbose        show verbose output
  -nc, --no-color      disable color in output
  -ls, --list-sources  list all available sources (-o for JSON)
  -stats              report source statistics

OPTIMIZATION:
  -timeout int          seconds to wait before timing out (default 30)
  -max-time int         minutes to wait for enumeration results (default 10)
  -rbr, --response-size-read int max response body size to read in bytes from passive sources (0 = unlimited)
```

8.7 Cross-Domain and Shared-Infrastructure Interpretation

Historical certificate data also contained cross-domain Common Name/name-set relationships. These were retained as attribution leads only. A shared certificate, hosting provider or infrastructure pattern is not sufficient to conclude ownership, compromise, takeover or misissuance.

Evidence 27 — Passive-recon evidence workspace

WHY WE PERFORMED THIS: Supporting capture for the broader PM4 evidence and correlation process.

WHAT THIS EVIDENCE SHOWS: Preserves the passive-reconnaissance workflow and source context used to make the namespace analysis reproducible.

```
(root@kali)~# ls
networkwalks_spiderfoot_passive.txt  networkwalks_w2pm1_dnsrecon.txt  networkwalks_w2pm1_whatweb.txt  networkwalks_w2pm1_wp-json.txt
networkwalks_w2pm1_curl_headers.txt  networkwalks_w2pm1_nslookup.txt  networkwalks_w2pm1_whois.txt

(root@kali)~# nano cURL_PM4

(root@kali)~# ls
cURL_PM4  networkwalks_spiderfoot_passive.txt  networkwalks_w2pm1_curl_headers.txt  networkwalks_w2pm1_nslookup.txt  networkwalks_w2pm1_whois.txt
networkwalks_w2pm1_whatweb.txt  networkwalks_w2pm1_wp-json.txt

(root@kali)~# nano cURL_PM4

(root@kali)~# ls
cURL_PM4  networkwalks_spiderfoot_passive.txt  networkwalks_w2pm1_curl_headers.txt  networkwalks_w2pm1_nslookup.txt  networkwalks_w2pm1_whois.txt
cURL_PM4.sh  networkwalks_w2pm1_curl_headers.txt  networkwalks_w2pm1_nslookup.txt  networkwalks_w2pm1_whois.txt

(root@kali)~# rm cURL_PM4

(root@kali)~# ls
cURL_PM4.sh  networkwalks_spiderfoot_passive.txt  networkwalks_w2pm1_curl_headers.txt  networkwalks_w2pm1_nslookup.txt  networkwalks_w2pm1_whois.txt
networkwalks_w2pm1_whatweb.txt  networkwalks_w2pm1_wp-json.txt

(root@kali)~# chmod +x cURL_PM4.sh

(root@kali)~# nano cURL_PM4.sh
```

9. W2-PM5 / PM-B — Nmap Network Scanning

PM5 moved the assessment from passive intelligence into controlled network enumeration. The objective was to resolve the target, establish reachability, map the TCP response surface, validate service identities, inspect common application and TLS surfaces, and preserve uncertainty where the evidence was inconclusive.

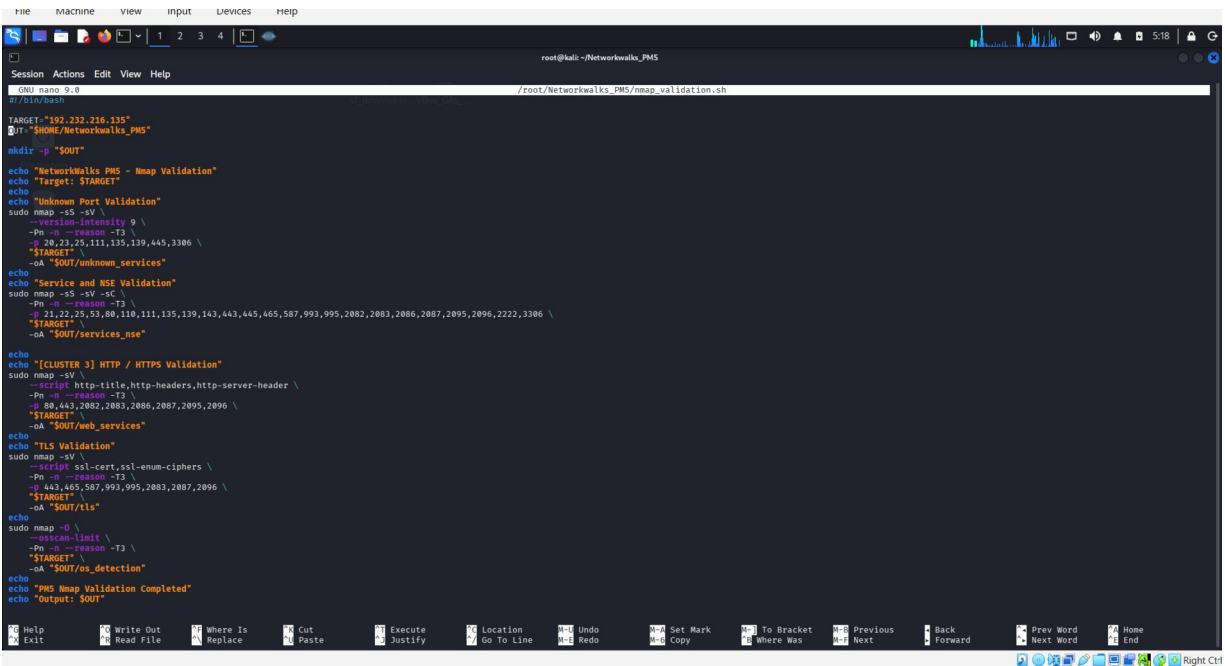
9.1 Nmap Environment and Interface Baseline

Nmap 7.99 was installed at /usr/bin/nmap. The assessment-side interface was eth0 with 10.0.0.2/24 and a default route through 10.0.0.1. This local network information is included for reproducibility and is not target infrastructure.

Evidence 28 — Nmap installation and capability baseline

WHY WE PERFORMED THIS: Confirms the Nmap build and available scanning capabilities.

WHAT THIS EVIDENCE SHOWS: Confirms the Nmap build, available capabilities and command options used for controlled network enumeration.



```
file machine view input devices help
root@kali:~/Networkwalks_PMS
Session Actions Edit View Help
root@kali:~/Networkwalks_PMS/mmap_validation.sh
# ./SOUT/nmap.sh
TARGET="192.232.216.135"
if [ "$HOME/Networkwalks_PMS" ]; then
  mkdir -p "$SOUT"
  echo "Networkwalks PMS - Nmap Validation"
  echo "Target: $TARGET"
  echo "Unknown Port Validation"
  sudo nmap -sS -sV \
    --version-intensity 9 \
    -p 20,22,25,111,135,139,445,3306 \
    "$TARGET" \
    -oA "$SOUT/unknown_services"
  echo "Service and OS Validation"
  sudo nmap -sS -sV -sC \
    -p 20,22,25,53,80,110,111,135,139,143,443,445,465,587,993,995,2082,2083,2086,2087,2095,2096,2222,3306 \
    "$TARGET" \
    -oA "$SOUT/services_nse"
  echo "[CLUSTER 3] HTTP / HTTPS Validation"
  sudo nmap -sV \
    --script http-title,http-headers,http-server-header \
    -p 80,443,2082,2083,2086,2087,2095,2096 \
    "$TARGET" \
    -oA "$SOUT/web_services"
  echo "TLS Validation"
  sudo nmap -sV \
    --script ssl-cert,ssl-enum-ciphers \
    -p 443,465,587,993,995,2083,2087,2096 \
    "$TARGET" \
    -oA "$SOUT/tls"
  echo "OS Detection"
  sudo nmap -O \
    --osscan-limit \
    -p 20,22,25,53,80,110,111,135,139,143,443,445,465,587,993,995,2082,2083,2086,2087,2095,2096 \
    "$TARGET" \
    -oA "$SOUT/os_detection"
  echo "PMS Nmap Validation Completed"
  echo "Output: $SOUT"
```

Evidence 29 — Nmap command / options evidence

WHY WE PERFORMED THIS: Supports reproducibility of the selected discovery and validation workflow.

WHAT THIS EVIDENCE SHOWS: Preserves the visual proof required to reproduce, validate and interpret this stage of the assessment.

```
--host-timeout <time>: Give up on target after this long
--scan-delay/--max-scan-delay <time>: Adjust delay between probes
--min-rate <number>: Send packets no slower than <number> per second
--max-rate <number>: Send packets no faster than <number> per second
FIREWALL/IDS EVASION AND SPOOFING:
-f; --mtu <val>: fragment packets (optionally w/given MTU)
-D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys
-S <IP_Address>: Spoof source address
-e <iface>: Use specified interface
-g/--source-port <portnum>: Use given port number
--proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies
--data <hex string>: Append a custom payload to sent packets
--data-string <string>: Append a custom ASCII string to sent packets
--data-length <num>: Append random data to sent packets
--ip-options <options>: Send packets with specified ip options
--ttl <val>: Set IP time-to-live field
--spooof-mac <mac address/prefix/vendor name>: Spoof your MAC address
--badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
-oN/-oX/-oS/-oG <file>: Output scan in normal, XML, s|<ri>pt kIdi3,
and Grepable format, respectively, to the given filename.
-oA <basename>: Output in the three major formats at once
-v: Increase verbosity level (use -vv or more for greater effect)
-d: Increase debugging level (use -dd or more for greater effect)
--reason: Display the reason a port is in a particular state
--open: Only show open (or possibly open) ports
--packet-trace: Show all packets sent and received
--iflist: Print host interfaces and routes (for debugging)
--append-output: Append to rather than clobber specified output files
--resume <filename>: Resume an aborted scan
--noninteractive: Disable runtime interactions via keyboard
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
--webxml: Reference stylesheet from Nmap.Org for more portable XML
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
-6: Enable IPv6 scanning
-A: Enable OS detection, version detection, script scanning, and traceroute
--datadir <dirname>: Specify custom Nmap data file location
--send-eth/--send-ip: Send using raw ethernet frames or IP packets
--privileged: Assume that the user is fully privileged
--unprivileged: Assume the user lacks raw socket privileges
```

Evidence 30 — Nmap environment evidence

WHY WE PERFORMED THIS: Additional baseline evidence from the assessment host.

WHAT THIS EVIDENCE SHOWS: Records the assessment host network context so scan results can be interpreted against the correct interface and route.

```

MISC:
  -G: Enable IPv6 scanning
  -A: Enable OS detection, version detection, script scanning, and traceroute
  --datadir <dirname>: Specify custom Nmap data file location
  --send-eth/--send-ip: Send using raw ethernet frames or IP packets
  --privileged: Assume that the user is fully privileged
  --unprivileged: Assume the user lacks raw socket privileges
  -V: Print version number
  -h: Print this help summary page.
EXAMPLES:
  nmap -v -A scanme.nmap.org
  nmap -v -sn 192.168.0.0/16 10.0.0.0/8
  nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES

(root@kali)~# zenmap --version
/usr/lib/python3/dist-packages/zenmapCore/I18N.py:77: DeprecationWarning: 'locale.getdefaultlocale' is deprecated and slated for removal in Python 3.15. Use setlocale(), getencoding() and g
etlocale() instead.
  loc, enc = locale.getdefaultlocale()
zenmap 7.99

(root@kali)~# which zenmap
/usr/bin/zenmap

```

Evidence 31 — Nmap interface / routing evidence

WHY WE PERFORMED THIS: Records the assessor-side interface and network context.

WHAT THIS EVIDENCE SHOWS: Records the assessment host network context so scan results can be interpreted against the correct interface and route.

```

(root@kali)~# mkdir -p ~/Networkwalks_PMS
(root@kali)~# nmap --version | tee ~/Networkwalks_PMS/PMS_nmap_version.txt
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(root@kali)~# nmap --iflist | tee ~/Networkwalks_PMS/PMS_nmap_interfaces.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2025-09-18 03:44 -0400
*****INTERFACES*****
DEV (SHORT) IP/MASK          TYPE UP MTU  MAC
lo (lo)      127.0.0.1/8      loopback up 65536
lo (lo)      ::1/128         loopback up 65536
eth0 (eth0)  10.0.0.2/24        ethernet up 1500  08:00:27:5A:87:BC
eth0 (eth0)  fe80::a10d:5f99:ff35:d935/64 ethernet up 1500  08:00:27:5A:87:BC

*****ROUTES*****
DST/MASK          DEV  METRIC GATEWAY
10.0.0.0/24       eth0 100    10.0.0.1
0.0.0.0/0         eth0 100    10.0.0.1
::1/128          lo    0
fe80::a10d:5f99:ff35:d935/128 eth0 0
fe80::/64        eth0 1024
ff00::/8         eth0 256

```

Evidence

Evidence 33 — Nmap capability evidence

WHY WE PERFORMED THIS: Supporting command output retained for the PM5 baseline.

WHAT THIS EVIDENCE SHOWS: Confirms the Nmap build, available capabilities and command options used for controlled network enumeration.

```
(root@kali)-[~]
# nmap --iflist
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 03:41 -0400
*****INTERFACES*****
DEV (SHORT) IP/MASK TYPE UP MTU MAC
lo (lo) 127.0.0.1/8 loopback up 65536
lo (lo) ::1/128 loopback up 65536
eth0 (eth0) 10.0.0.2/24 ethernet up 1500 08:00:27:5A:87:BC
eth0 (eth0) fe80::a10d:5f99:ff35:d935/64 ethernet up 1500 08:00:27:5A:87:BC

*****ROUTES*****
DST/MASK DEV METRIC GATEWAY
10.0.0.0/24 eth0 100
0.0.0.0/0 eth0 100 10.0.0.1
::1/128 lo 0
fe80::a10d:5f99:ff35:d935/128 eth0 0
fe80::/64 eth0 1024
ff00::/8 eth0 256
```

Evidence 35 — Nmap execution environment

WHY WE PERFORMED THIS: Additional evidence from the Nmap setup and execution sequence.

WHAT THIS EVIDENCE SHOWS: Preserves the visual proof required to reproduce, validate and interpret this stage of the assessment.

```
(root@kali)-[~]
# nmap --version
Nmap version 7.99 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcr2-10.46 libpcap-1.10.6 nmap-libidn2-1.18.0 ipv6
Compiled without:
Available nsock engines: epoll poll select

(root@kali)-[~]
# which nmap
/usr/bin/nmap

(root@kali)-[~]
# nmap -h
Nmap 7.99 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
```

9.2 Target Resolution and Host Discovery

Nmap resolved networkwalks.com to 192.232.216.135 and displayed the reverse DNS name 192-232-216-135.unifiedlayer.com. Host discovery reported the target as up.

Evidence 34 — Nmap target resolution and host discovery

WHY WE PERFORMED THIS: Primary evidence for the target IP resolution and reachability result.

WHAT THIS EVIDENCE SHOWS: Confirms the target address used by Nmap and establishes that the host responded during discovery.

```
(root@kali)-[~]
# nmap -sL networkwalks.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 03:46 -0400
Nmap scan report for networkwalks.com (192.232.216.135)
rDNS record for 192.232.216.135: 192-232-216-135.unifiedlayer.com
Nmap done: 1 IP address (0 hosts up) scanned in 1.35 seconds

(root@kali)-[~]
# nmap -sL networkwalks.com | tee ~/Networkwalks_PM5/PM5_target_resolution.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 03:46 -0400
Nmap scan report for networkwalks.com (192.232.216.135)
rDNS record for 192.232.216.135: 192-232-216-135.unifiedlayer.com
Nmap done: 1 IP address (0 hosts up) scanned in 0.94 seconds

(root@kali)-[~]
# nmap -sn networkwalks.com
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 03:47 -0400
Nmap scan report for networkwalks.com (192.232.216.135)
Host is up (0.00082s latency).
rDNS record for 192.232.216.135: 192-232-216-135.unifiedlayer.com
Nmap done: 1 IP address (1 host up) scanned in 1.20 seconds

(root@kali)-[~]
# nmap -sn networkwalks.com | tee ~/Networkwalks_PM5/PM5_host_discovery.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 03:47 -0400
Nmap scan report for networkwalks.com (192.232.216.135)
Host is up (0.00052s latency).
rDNS record for 192.232.216.135: 192-232-216-135.unifiedlayer.com
Nmap done: 1 IP address (1 host up) scanned in 1.21 seconds
```

9.3 Broad TCP Discovery

A full 1–65535 TCP attempt became impractical because response timing varied substantially. The completed top-10,000 TCP scan then reported 3,410 open TCP states and 4,977 filtered ports, alongside retransmission-cap and RTTVAR warnings.

The unusually large open-state population is treated as a response-surface observation rather than a count of thousands of distinct applications. Targeted service validation provides the stronger attribution layer.

Evidence 38 — Top-10,000 TCP scan

WHY WE PERFORMED THIS: Primary evidence for the broad TCP discovery phase and scan timing behavior.

WHAT THIS EVIDENCE SHOWS: Documents broad TCP enumeration and scan behavior; candidate states were treated as observations requiring targeted validation.

```

root@kali: ~/Networkwalks_PMS
Session Actions Edit View Help
54045/tcp open unknown syn-ack ttl 64
55056/tcp open unknown syn-ack ttl 64
55555/tcp open unknown syn-ack ttl 64
55600/tcp open unknown syn-ack ttl 64
56738/tcp open unknown syn-ack ttl 64
57797/tcp open unknown syn-ack ttl 64
58080/tcp open unknown syn-ack ttl 64
60020/tcp open unknown syn-ack ttl 64
60443/tcp open unknown syn-ack ttl 64
61900/tcp open unknown syn-ack ttl 64
62078/tcp open iphone-sync syn-ack ttl 64
63331/tcp open unknown syn-ack ttl 64
64623/tcp open unknown syn-ack ttl 64
64680/tcp open unknown syn-ack ttl 64
65000/tcp open unknown syn-ack ttl 64
65129/tcp open unknown syn-ack ttl 64
65389/tcp open unknown syn-ack ttl 64
Device type: VoIP adapter/bridge/general purpose
Running (JUST GUESSING): AT&T embedded (90%), Oracle Virtualbox (89%), Slirp (89%), QEMU (89%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny.gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: AT&T BGW210 voice gateway (90%), Oracle Virtualbox Slirp NAT bridge (89%), QEMU user mode network gateway (89%)
No exact OS matches for host (test conditions non-ideal).

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 89.62 seconds

PMS Nmap Validation Completed
Output: /root/Networkwalks_PMS

(root@kali) ~/Networkwalks_PMS
ls
nmap_validation.sh PMS_full_tcp.gnmap PMS_nmap_interfaces.txt PMS_service_validation.xml PMS_tcp_top10000.xml tls.gnmap unknown_services.nmap web_services.xml
os_detection.gnmap PMS_full_tcp.nmap PMS_nmap_version.txt PMS_target_resolution.txt services_nse.gnmap tls.nmap unknown_services.xml
os_detection.nmap PMS_full_tcp.xml PMS_service_validation.gnmap PMS_tcp_top10000.gnmap services_nse.nmap tls.xml web_services.gnmap
os_detection.xml PMS_host_discovery.txt PMS_service_validation.nmap PMS_tcp_top10000.nmap services_nse.xml unknown_services.gnmap web_services.nmap

(root@kali) ~/Networkwalks_PMS
nano ~/Networkwalks_PMS/nmap_validation.sh

```

Evidence 39 — TCP open-state extraction

WHY WE PERFORMED THIS: Supports the extraction of candidate open TCP states from the broad scan.

WHAT THIS EVIDENCE SHOWS: Documents broad TCP enumeration and scan behavior; candidate states were treated as observations requiring targeted validation.

```

(root@kali) ~
$ sudo nmap -sS --top-ports 10000 -Pn -n --reason -T4 --stats-every 15s 192.232.216.135 -oA ~/Networkwalks_PMS/PMS_tcp_top10000
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 04:02 -0400
Stats: 0:00:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 3.52% done; ETC: 04:10 (0:06:51 remaining)
Stats: 0:00:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 6.84% done; ETC: 04:10 (0:06:49 remaining)
Stats: 0:00:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 10.58% done; ETC: 04:10 (0:06:20 remaining)
Stats: 0:01:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 13.53% done; ETC: 04:10 (0:06:23 remaining)
Stats: 0:01:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 16.79% done; ETC: 04:10 (0:06:12 remaining)
Stats: 0:01:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 19.88% done; ETC: 04:10 (0:06:03 remaining)
Stats: 0:01:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 22.84% done; ETC: 04:10 (0:05:55 remaining)
Stats: 0:02:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 25.81% done; ETC: 04:10 (0:05:45 remaining)
Stats: 0:02:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 28.60% done; ETC: 04:10 (0:05:37 remaining)
Stats: 0:02:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 31.77% done; ETC: 04:10 (0:05:22 remaining)
Stats: 0:02:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 35.15% done; ETC: 04:10 (0:05:04 remaining)
Warning: 192.232.216.135 giving up on port because retransmission cap hit (6).
Stats: 0:03:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 35.61% done; ETC: 04:11 (0:05:25 remaining)
Stats: 0:03:15 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 36.12% done; ETC: 04:11 (0:05:45 remaining)
Stats: 0:03:30 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 38.10% done; ETC: 04:12 (0:05:41 remaining)
Stats: 0:03:45 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 40.13% done; ETC: 04:12 (0:05:36 remaining)
Stats: 0:04:00 elapsed; 0 hosts completed (1 up), 1 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 42.27% done; ETC: 04:12 (0:05:28 remaining)

```


9.4 Service and Version Validation

Targeted validation positively identified the following services:

Port	Validated service / version	Interpretation
21/tcp	Pure-FTPd	FTP service externally reachable
22/tcp	OpenSSH 9.9	SSH service
53/tcp	ISC BIND 9.16.23	DNS service
80/tcp	Apache HTTP Server	HTTP service
110/tcp	Dovecot POP3	Mail retrieval service
143/tcp	Dovecot IMAP	Mail retrieval service
443/tcp	Apache HTTP Server	HTTPS service
465/tcp	Exim smtpd 4.100	SMTPS submission/mail service
587/tcp	Exim smtpd 4.100	SMTP submission service
993/tcp	Dovecot IMAPS	Encrypted IMAP
995/tcp	Dovecot POP3S	Encrypted POP3
2222/tcp	OpenSSH 9.9	Second SSH listener

Evidence 32 — Service validation output

WHY WE PERFORMED THIS: Primary evidence for validated service identities.

WHAT THIS EVIDENCE SHOWS: Provides protocol/service-level evidence used to distinguish validated services from scanner hypotheses and candidate states.

```
(root@kali) ~ /Networkwalks_PMS
# ./nmap_validation.sh
NetworkWalks PMS - Nmap Validation
Target: 192.232.216.135

Unknown Port Validation
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 04:43 -0400
Stats: 0:05:07 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 04:54 (0:03:06 remaining)
Stats: 0:05:14 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 04:54 (0:05:12 remaining)
Stats: 0:05:47 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 04:55 (0:03:45 remaining)
Stats: 0:06:37 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 04:57 (0:06:35 remaining)
Stats: 0:06:38 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 04:57 (0:06:36 remaining)
Stats: 0:08:43 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 50.00% done; ETC: 05:01 (0:08:41 remaining)
Stats: 0:10:00 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.72% done; ETC: 04:53 (0:00:00 remaining)
Nmap scan report for 192.232.216.135
Host is up, received user-set (0.54s latency).

PORT      STATE SERVICE      REASON      VERSION
20/tcp    open  tcprapped    syn-ack ttl 64
23/tcp    open  tcprapped    syn-ack ttl 64
25/tcp    open  smtp?        syn-ack ttl 64
111/tcp   open  tcprapped    syn-ack ttl 64
135/tcp   open  tcprapped    syn-ack ttl 64
139/tcp   open  netbios-ssn? syn-ack ttl 64
445/tcp   open  microsoft-ds? syn-ack ttl 64
3306/tcp  open  mysql?       syn-ack ttl 64

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 607.57 seconds

Service and NSE Validation
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 04:54 -0400
Stats: 0:00:55 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 42.86% done; ETC: 04:56 (0:01:11 remaining)
Stats: 0:01:10 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 85.71% done; ETC: 04:55 (0:00:12 remaining)
Stats: 0:01:48 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 85.71% done; ETC: 04:56 (0:00:18 remaining)
```

Evidence 36 — Targeted version validation

WHY WE PERFORMED THIS: Additional service/version evidence.

WHAT THIS EVIDENCE SHOWS: Provides protocol/service-level evidence used to distinguish validated services from scanner hypotheses and candidate states.

```
(root@kali) ~
# nano ~/Networkwalks_PMS/nmap_validation.sh
# cd Networkwalks_PMS
ls
nmap_validation.sh  PMS_full_tcp.nmap  PMS_host_discovery.txt  PMS_nmap_version.txt  PMS_service_validation.nmap  PMS_target_resolution.txt  PMS_tcp_top10000.nmap
PMS_full_tcp.gnmap  PMS_full_tcp.xml  PMS_nmap_interfaces.txt  PMS_service_validation.gnmap  PMS_service_validation.xml  PMS_tcp_top10000.gnmap  PMS_tcp_top10000.xml
# chmod +x Networkwalks_PMS/nmap_validation.sh
chmod: cannot access 'Networkwalks_PMS/nmap_validation.sh': No such file or directory
# chmod +x nmap_validation.sh
```

Evidence 37 — Service validation / protocol evidence

WHY WE PERFORMED THIS: Supporting capture for validated network services.

WHAT THIS EVIDENCE SHOWS: Provides protocol/service-level evidence used to distinguish validated services from scanner hypotheses and candidate states.

```
(root@kali)~#
# sudo nmap -sS -sV --version-intensity 5 -Pn -n --reason -T3 -p 20,21,22,23,25,53,80,110,111,135,139,143,443,445,465,587,993,995,2082,2083,2086,2087,2095,2096,2222,3306 192.232.216.135 -
oh ~/Networkwalks: PM5/PM5_service_validation
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-18 04:21 -0400
Stats: 0:01:55 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 92.31% done; ETC: 04:23 (0:00:09 remaining)
Nmap scan report for 192.232.216.135
Host is up, received user-set (1.2s latency).

PORT      STATE SERVICE      REASON      VERSION
20/tcp    open  tcpwrapped   syn-ack ttl 64
21/tcp    open  ftp          syn-ack ttl 64 Pure-FTPd
22/tcp    open  ssh          syn-ack ttl 64 OpenSSH 9.9 (protocol 2.0)
23/tcp    open  tcpwrapped   syn-ack ttl 64
25/tcp    open  smtp?        syn-ack ttl 64
53/tcp    open  domain       syn-ack ttl 64 ISC BIND 9.16.23 (RedHat Linux)
80/tcp    open  http         syn-ack ttl 64 Apache httpd
110/tcp   open  pop3         syn-ack ttl 64 Dovecot pop3d
111/tcp   open  tcpwrapped   syn-ack ttl 64
135/tcp   open  tcpwrapped   syn-ack ttl 64
139/tcp   open  netbios-ssn? syn-ack ttl 64
143/tcp   open  imap         syn-ack ttl 64 Dovecot imapd
443/tcp   open  ssl/http     syn-ack ttl 64 Apache httpd
445/tcp   open  microsoft-ds? syn-ack ttl 64
465/tcp   open  ssl/smtp     syn-ack ttl 64 Exim smtpd 4.100
587/tcp   open  smtp         syn-ack ttl 64 Exim smtpd 4.100
993/tcp   open  ssl/imap     syn-ack ttl 64 Dovecot imapd
995/tcp   open  ssl/pop3     syn-ack ttl 64 Dovecot pop3d
2082/tcp  open  infowave?    syn-ack ttl 64
2083/tcp  open  ssl/radsec?   syn-ack ttl 64
2086/tcp  open  gnuenet?     syn-ack ttl 64
2087/tcp  open  ssl/eli?     syn-ack ttl 64
2095/tcp  open  nbx-ser?     syn-ack ttl 64
2096/tcp  open  ssl/nbx-dir? syn-ack ttl 64
2222/tcp  open  ssh          syn-ack ttl 64 OpenSSH 9.9 (protocol 2.0)
3306/tcp  open  mysql?       syn-ack ttl 64

6 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====
NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)
=====
SF-Port2082-TCP:V=7.99%I=5ND=9/18&TIme=6AACF4ECMP=x86_64-pc-linux-gnu&r(SI
SF:Options,177F,*HTTP/1.0\\x20000\\x20Not\\x20Found\\r\\nContent-Type:\\x20tex
=====
```

Evidence 42 — NSE and service evidence

WHY WE PERFORMED THIS: Additional evidence for protocol-level validation.

WHAT THIS EVIDENCE SHOWS: Provides protocol/service-level evidence used to distinguish validated services from scanner hypotheses and candidate states.

```
-b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
-p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
--exclude-ports <port ranges>: Exclude the specified ports from scanning
-P: Fast mode - Scan fewer ports than the default scan
-P: Scan ports sequentially - don't randomize
--top-ports <number>: Scan <number> most common ports
--port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
-sv: Probe open ports to determine service/version info
--version-intensity <level>: Set from 0 (light) to 9 (try all probes)
--version-light: Limit to most likely probes (intensity 2)
--version-all: Try every single probe (intensity 9)
--version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
-sc: equivalent to --script-default
--script=<Lua scripts>: <Lua scripts> is a comma separated list of
  directories, script-files or script-categories
--script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts
--script-args-file=filename: provide NSE script args in a file
--script-trace: Show all data sent and received
--script-updatedb: Update the script database.
--script-help=<lua scripts>: Show help about scripts.
  <lua scripts> is a comma-separated list of script-files or
  script-categories.
OS DETECTION:
-O: Enable OS detection
--osscan-limit: Limit OS detection to promising targets
--osscan-guess: Guess OS more aggressively
TIMING AND PERFORMANCE:
Options which take <time> are in seconds, or append 'ms' (milliseconds),
's' (seconds), 'm' (minutes), or 'h' (hours) to the value (e.g. 30m).
-T<0-5>: Set timing template (higher is faster)
--min-hostgroup/max-hostgroup <size>: Parallel host scan group sizes
--min-parallelism/max-parallelism <numprobes>: Probe parallelization
--min-rtt-timeout/max-rtt-timeout/initial-rtt-timeout <time>: Specifies
  probe round trip time.
--max-retries <tries>: Caps number of port scan probe retransmissions.
--host-timeout <time>: Give up on target after this long
--scan-delay/--max-scan-delay <time>: Adjust delay between probes
```

9.5 Candidate and Unresolved Services

Several ports produced candidate labels such as tcpwrapped, smtp?, netbios-ssn?, microsoft-ds?, mysql? or other service guesses. These labels were not converted into confirmed applications because higher-intensity validation did not provide sufficiently strong protocol evidence.

Analyst lesson

Nmap's service database can provide useful hypotheses, but professional reporting requires protocol-level evidence before calling an application confirmed.

9.6 HTTP, Hosting-Management and Webmail Surfaces

HTTP validation identified Apache on the main web ports and application-specific response patterns consistent with cPanel-style, WHM-style and webmail interfaces on ports 2083, 2087 and 2096. These are externally reachable surfaces that an authorized tester would prioritize for configuration and authentication review.

The evidence does not establish an authorization flaw, credential weakness or exploitable application vulnerability on those interfaces.

9.7 NSE, Mail Protocols and TLS

NSE and protocol validation exposed POP3/IMAP capabilities and SMTP STARTTLS/AUTH behavior. TLS inspection showed TLS 1.2/1.3 support and provider-associated certificates, including *.hostgator.com on several management/mail surfaces. These observations are useful for configuration review but do not, by themselves, establish a cryptographic vulnerability or ownership relationship.

9.8 OS Detection

OS fingerprinting produced several environment/device guesses, including VirtualBox/Slirp/QEMU/embedded possibilities, but Nmap explicitly reported that no exact OS match was available under the test conditions. The guesses are therefore retained as low-confidence fingerprinting output rather than as confirmed operating-system identification.

10. Zenmap — Graphical Nmap Workflow

Zenmap 7.99 provides a graphical interface to Nmap. It was used to present the Nmap workflow through target/profile configuration and graphical result-oriented views. The GUI complements the command-line evidence rather than replacing the reproducible Nmap outputs.

10.1 Zenmap Configuration

The captured GUI configuration shows target 192.232.216.135 with the Intense scan profile and the generated Nmap command. This directly demonstrates the graphical configuration layer.

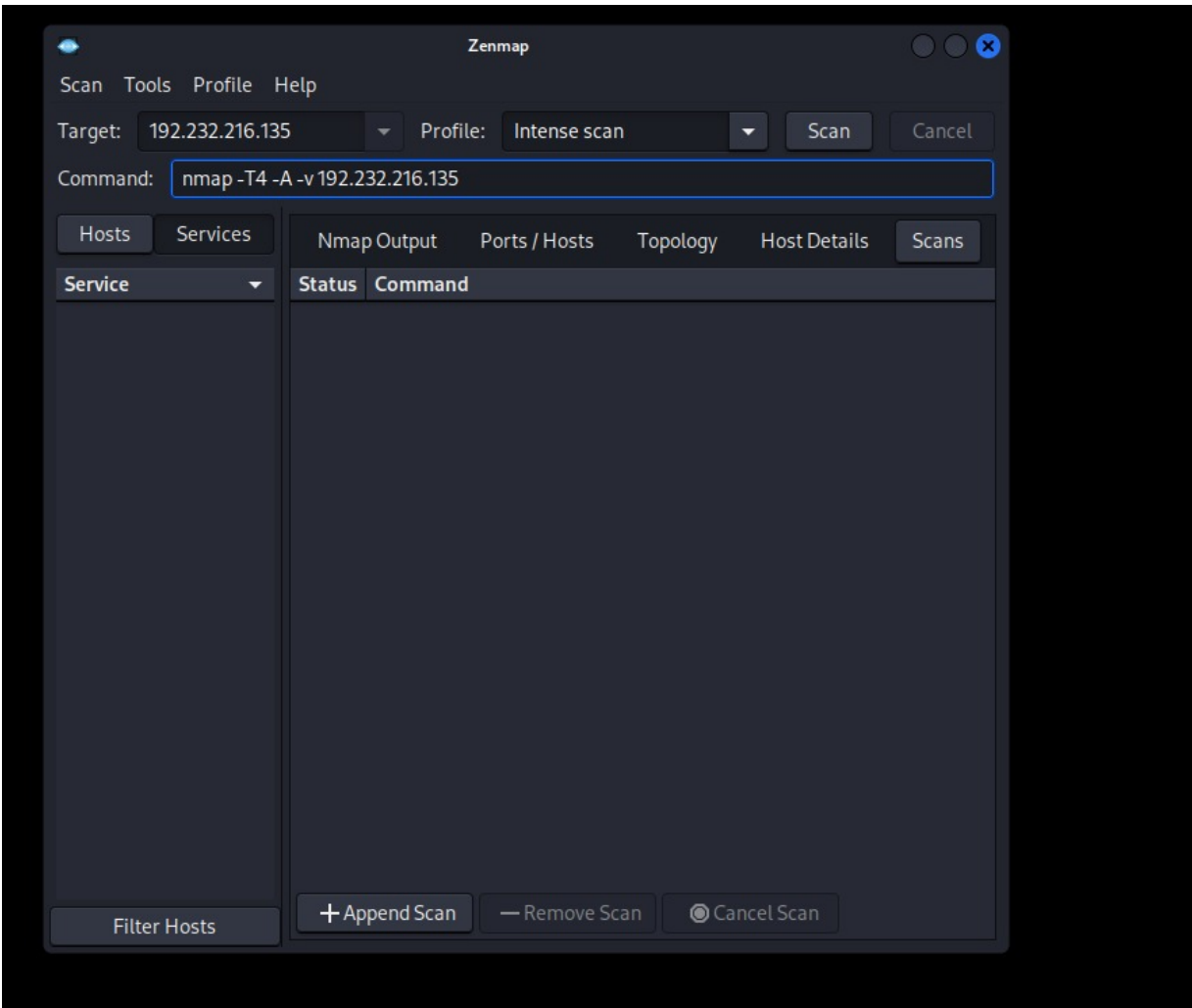
10.2 Zenmap and Nmap Correlation

The preserved Zenmap source-evidence montage documents the associated Nmap/Zenmap workflow material retained with the original PM5 evidence archive. It is used as supplementary source evidence alongside the direct GUI capture.

Evidence 49 — Earlier Zenmap GUI evidence

WHY WE PERFORMED THIS: Additional GUI capture retained from the PM5 evidence set.

WHAT THIS EVIDENCE SHOWS: Shows the graphical Nmap workflow, target/profile configuration and correlation with the command-line assessment evidence.



Evidence 52 — Zenmap source-evidence montage

WHY WE PERFORMED THIS: Supplementary visual record from the preserved Zenmap/Nmap evidence set; used to maintain the full evidence trail.

WHAT THIS EVIDENCE SHOWS: Shows the graphical Nmap workflow, target/profile configuration and correlation with the command-line assessment evidence.



Nmap_Zenmap1.png



Nmap_Zenmap3.png

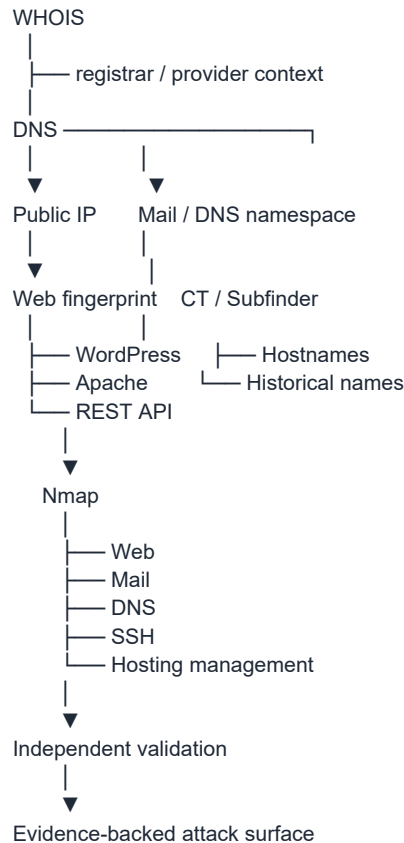
Nmap_Zenmap2.png



Nmap_Zenmap4.png

The practical value of Zenmap is that it makes scan configuration and result review accessible through a graphical workflow while the command-line outputs remain the authoritative reproducibility layer.

11. Cross-Module Correlation and Attacker Perspective



An attacker would not normally treat these tools as isolated exercises. The useful chain is to use DNS and CT to identify candidate names, use web fingerprinting to identify application technologies, use Nmap to understand exposed network services, and then validate whether the relationships represent the same current infrastructure.

The most important correlation in PM4 is the eight-name overlap between Subfinder and the ten-name CT-derived namespace. The most important correlation in PM5 is between the broad TCP response surface and targeted service validation: the latter prevents the broad open-state count from being misreported as thousands of independent services.

12. Findings, Observations and Non-Findings

The table below distinguishes what was directly observed from what would require additional validation.

Item	Evidence-backed observation	Potential security relevance	Validation needed
Public IP	networkwalks.com → 192.232.216.135	Provides a direct network-service target	Ownership/current infrastructure context
DNS provider	HostGator nameservers observed	Provider correlation and infrastructure context	Do not infer ownership of other provider assets
Web stack	Apache + WordPress-related fingerprints	Enables technology-specific review	Independent version/component verification
REST API reference	WordPress REST API Link header observed	Creates an application enumeration pivot	Authorized endpoint/content access review
WAF	ModSecurity (SpiderLabs) fingerprint	Indicates a defensive layer	Coverage, rules and bypass resistance testing
CT namespace	10 normalized certificate-derived names	Expands passive asset inventory	Current DNS/addressability checks
Subfinder overlap	8 CT names also discovered by Subfinder	Cross-source confidence	Current liveness and ownership

Broad TCP response	3,410 open states / 4,977 filtered in top-10k	Large response surface	Repeatability and service-level attribution
Mail services	POP3/IMAP/SMTP and encrypted variants validated	Externally reachable authentication surfaces	Configuration/authentication review
SSH	OpenSSH 9.9 on 22 and 2222	Remote administration surface	Authorized authentication/configuration review
Hosting management	cPanel/WHM/webmail-style responses	Administrative interfaces exposed	Authorized authentication and access-control testing
Confirmed CVE	None established by these phases	Avoids unsupported vulnerability claims	Separate authorized vulnerability validation

12.1 Explicit Non-Findings

- No credential compromise was demonstrated.
- No anonymous FTP access was demonstrated.
- No open SMTP relay was demonstrated.
- No DNS zone-transfer weakness was demonstrated.
- No SSH authentication weakness was demonstrated.
- No cPanel/WHM authorization bypass was demonstrated.
- No MySQL authentication weakness was demonstrated.
- No confirmed CVE or exploitable vulnerability was established by the captured reconnaissance/enumeration evidence.
- No unauthorized access or persistence was demonstrated.

13. Security Recommendations

Priority area	Recommended action	Reason
Web technology exposure	Maintain accurate inventory and patch CMS/plugins; verify versions independently	Reduces risk from known component weaknesses and stale technology
HTTP security metadata	Review response headers and remove unnecessary implementation detail where practical	Limits passive fingerprinting and improves browser security posture
REST API	Review publicly reachable namespaces/routes and confirm intended information disclosure	Ensures public interfaces expose only intended data
DNS	Periodically review A/AAAA/CNAME/MX/TXT/NS/SOA records	Removes stale records and reduces unintended exposure
Certificate inventory	Monitor Certificate Transparency for unexpected hostnames	Provides an external view of certificate issuance and namespace drift
Subdomain inventory	Correlate passive discovery with authoritative DNS and asset ownership	Prevents stale/historical names from becoming blind spots
SSH	Review authentication policy, exposed listeners and management access controls	Reduces risk on remote administration surfaces
Mail	Review TLS, authentication, relay policy and service exposure	Protects externally reachable mail services
cPanel/WHM/webmail	Restrict management interfaces where feasible and enforce strong authentication/access controls	Administrative interfaces have higher operational impact if compromised
WAF	Keep ModSecurity maintained and monitor/tune rules and logs	A WAF is most useful when coverage and monitoring are validated
Network exposure	Reconcile externally reachable services against the intended service inventory	Removes unnecessary listening services

14. Beginner-Friendly Explanation of the Assessment

A simple way to understand the Week 2 workflow is to imagine that a security analyst is building a map before checking whether any door is actually unlocked.

Step	Simple meaning	Professional meaning
1. WHOIS	Who manages the domain and where is DNS delegated?	Administrative and delegation context
2. DNS	Where does the name point and what other records exist?	Infrastructure and service mapping
3. WhatWeb	What technology does the website appear to	Technology fingerprinting

	use?	
4. cURL	What does the server reveal in its response?	Header, cookie and interface metadata
5. WAFW00F	Is a protective layer visible?	Defensive architecture fingerprint
6. GHDB	What might search engines have indexed?	Search-engine footprinting
7. Maltego	Which entities appear related?	Relationship-oriented intelligence
8. CT / Subfinder	What names have appeared in certificates or passive sources?	Historical/current namespace intelligence
9. Nmap	Which network doors respond?	Port and service enumeration
10. Service validation	What is actually behind those responding ports?	Protocol/version attribution
11. Zenmap	Can the Nmap workflow be reviewed graphically?	GUI-based scan configuration and analysis
12. Reporting	What do the observations actually prove?	Evidence, attribution, impact and remediation

Core lesson

Finding an exposed service is not the same as finding a vulnerability. The analyst's job is to move from signal → validation → attribution → security significance.

15. Evidence Index — All Captured Screenshots

Every assembled evidence capture is reproduced in this report. The evidence number is used consistently throughout the document so a reviewer can move from a technical statement to the corresponding visual proof.

Evidence ID / Description	Evidence ID / Description
01 — Initial DNS / target resolution context	02 — WAFW00F detection
03 — WhatWeb web technology fingerprint	04 — PM1 WHOIS / evidence preservation
05 — DNS record enumeration	06 — DNS validation output
07 — DNS infrastructure / record evidence	08 — Supplementary PM1 capture
09 — DNS resolution validation	10 — Additional WHOIS / PM1 capture
11 — dig command reference	12 — whois command reference
13 — DNS validation capture	14 — DNSRecon command reference
15 — DNS resolution validation	16 — cURL HTTP headers
17 — DNSRecon timeout result	18 — DNSRecon explicit resolver attempt
19 — theHarvester / CertSpotter baseline	20 — theHarvester / reconnaissance tooling context
21 — Direct Certificate Transparency collection	22 — crt.sh adapter result
23 — PM4 passive-recon tool context	24 — PM4 certificate-source workflow
25 — Historical / certificate names	26 — Certificate timeline evidence
27 — Passive-recon evidence workspace	28 — Nmap installation baseline
29 — Nmap options and capability output	30 — Nmap execution environment
31 — Nmap interface / routing context	32 — Service validation output
33 — Nmap capability evidence	34 — Nmap target resolution and host discovery
35 — Nmap execution environment evidence	36 — Targeted service / HTTP validation
37 — Service validation evidence	38 — Top-10,000 TCP scan
39 — TCP open-state extraction	40 — Broad TCP response surface
41 — TCP timing / state behavior	42 — NSE / protocol evidence
43 — Additional DNSRecon evidence	44 — Week 2 project-module reference
45 — Additional cURL HTTP evidence	46 — Subfinder passive discovery
47 — WHOIS registration evidence	48 — Subfinder baseline
49 — Zenmap GUI evidence	50 — Repeated DNSRecon evidence
51 — Subfinder methodology / output	
52 — Zenmap source-evidence montage	

Evidence placement rule: each numbered evidence item is paired with its corresponding screenshot directly beneath the purpose and interpretation fields. Duplicate screenshot placements have been removed so each unique capture is presented once in its relevant assessment section.

16. Final Conclusion

The Week 2 assessment produced a structured, evidence-backed view of the Networkwalks external attack surface. PM1 established the public DNS, web and defensive-architecture baseline. PM2 introduced search-engine footprinting as an intelligence source. PM3 supplied the relationship-analysis model needed to connect domains, certificates, hosts and infrastructure. PM4 expanded the namespace through passive certificate and subdomain intelligence. PM5 then moved into controlled network enumeration and service validation, with Zenmap providing the graphical Nmap workflow.

The strongest technical lesson is methodological: raw tool output should be treated as a measurement, not a verdict. Direct DNS validation strengthened the interpretation of DNSRecon behavior; Subfinder strengthened the CT-derived namespace through independent overlap; and targeted service detection prevented the broad Nmap open-state population from being mistaken for thousands of confirmed applications.

The resulting evidence is suitable as a professional Week 2 internship submission because it explains not only what commands were run, but why they were selected, what the results mean from an attacker and defender perspective, what evidence supports each observation, and where additional authorized validation would be required before calling something a confirmed vulnerability.

Final assessment statement

No confirmed high-impact vulnerability is asserted from the reconnaissance and enumeration evidence contained in this report. The report intentionally preserves uncertainty where the evidence is insufficient and identifies concrete attack-surface areas for subsequent authorized security validation.

17. References and Preserved Source Material

- Networkwalks W2-PM2 LAB PRACTICE.pdf — supplied GHDB practical/module instructions.
- Networkwalks W2-PM-FINAL Sample Report v2 — supplied report structure/reference.
- Networkwalks W2-PM1 Comprehensive Reconnaissance Report — PM1 command interpretation and evidence context.
- Networkwalks W2-PM4 Passive Reconnaissance Report v2 — certificate intelligence and Subfinder correlation.
- Networkwalks W2-PM5 Final Nmap Reports — Nmap command register, service validation, NSE, HTTP/TLS and OS interpretation.
- Preserved raw Nmap outputs: PM5_service_validation.nmap, PM5_tcp_top10000.nmap, os_detection.nmap, services_nse.nmap, tls.nmap, unknown_services.nmap, web_services.nmap.
- Preserved Week 2 screenshot evidence set: Evidence 01–52 (unique captures after duplicate removal).